IHP SG13G2 · OPEN SOURCE RFIC EM WORKFLOW

gds2palace FEM workflow for IHP

User's Guide

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1 What's New

This chapter gives a brief overview of major features added since the previous edition of this guide. For the complete, dated change log, see `CHANGES.md` in the repository.

- **settings['conductor_model']** chooses the conductor model in one setting, with names as in HFSS: `'surface'` (default, surface impedance), `'solve_inside'` (conductors meshed as volumes) or `'solve_inside_skin'` (volumes with thin layers along their faces for the skin effect). With the surface model, the Q of coils on TopMetal1/TopMetal2 is 18-30 % too high; for Q and loss, use `'solve_inside'` while designing and `'solve_inside_skin'` for final numbers. See "Choosing the conductor model" in chapter "Conductor loss modelling".
- Expert settings behind `conductor_model`: **settings['filled_metals']** (conductors as volumes: `True` for all conductor layers, `'auto'` for the layers thinner than `refined_cellsize` only) and **settings['skin_layers']** (automatic depths, a list of depths, or per layer). See "Conductor modelling" in chapter "settings".
- **settings['z_thickness_factor'] was removed.** The reduced side wall thickness also changed the side wall impedance well into the skin effect regime, and was not closer to measurement than the full thickness. Scripts that still set it get a warning, and the full thickness is used.
- **Elmer FEM as an additional solver.** gds2palace can now generate simulation input not only for AWS Palace, but also for Elmer FEM, an open-source multiphysics solver. This adds two new workflows, both built from the same GDSII layout + XML stackup + Python model script approach already used for Palace:
  - **Elmer EM simulation** — S-parameter simulation, see chapter "Using gds2palace with Elmer FEM for EM simulation".
  - **Elmer thermal simulation** — steady-state heat conduction (temperature) simulation from user-defined heat sources and constant-temperature boundaries, see chapter "Using gds2palace with Elmer FEM for thermal simulation".
- **Derived layers in the XML stackup.** A layer's geometry can now be computed from other layers using boolean operations (AND/OR/XOR/NOT) and resizing, instead of only being read directly from GDSII. This is used, for example, to derive SG13G2 resistor geometry from existing process layers, see the example in folder `more_examples/derived_layers_and_resistors`.
- **Reference-relative layer positioning in the XML stackup.** A `<Layer>` or `<Dielectric>` can now be positioned as an offset from the top or bottom edge of another named layer/dielectric, instead of always requiring an absolute Zmin/Zmax. This removes the need to hand-recompute every dependent layer's z-position whenever a thickness changes.
- **Simplified handling of cutouts.** Mesh generation was redesigned so that the `preprocess_gds` option is no longer required for layouts with cutouts (holes) or other self-intersecting polygon boundaries.
- **setupThermal.** The companion desktop tool `setupEM` now also includes `setupThermal`, a GUI for building Elmer thermal models without writing Python code, and a GUI-driven XML stackup editor. See https://github.com/VolkerMuehlhaus/setupEM

The XML stackup file format used by both Palace and Elmer models has grown to support these new features. See chapter "Extended XML stackup format" in the Appendix, and the full attribute reference in `XML_stackup_format.md`.

2 About this workflow

Palace, for **PA** rallel **LA** rge-scale **C** omputational **E** lectromagnetics, is an open-source, parallel finite element code for full-wave 3D electromagnetic simulations. It can be scaled from single computer to large high performance simulation clusters and cloud-based computing.
https://awslabs.github.io/palace/stable/
https://aws.amazon.com/de/blogs/quantum-computing/aws-releases-open-source-software-palace-for-

cloud-based-electromagnetics-simulations-of-quantum-computing-hardware/
The gds2palace workflow enables RFIC FEM simulation using Palace from GDSII layout files.

3 Workflow

In this document, the status of the IHP workflow developed for AWS Palace FEM solver is documented.
Two files must be provided by the user: the **layout in GDSII format** and a **simulation model script** in Python. This simulation script references the technology stackup file (XML format) and calls functions from the workflow code to create two output files for Palace: the **simulation mesh file** which is built from geometries and stackup, and the **simulation control file** in *.json format which defines material settings and simulation settings.

Having these two files, the user can now run the FEM simulation in Palace. The simulation model script can start an external command to start simulation, or the user can run Palace on a platform of his choice. This way, the files for Palace can be created on any desktop computer, and the Palace simulation can be done on the same computer or on a very different system. This gives many options to scale Palace simulation power as needed, using the exact same Palace input data from our workflow.

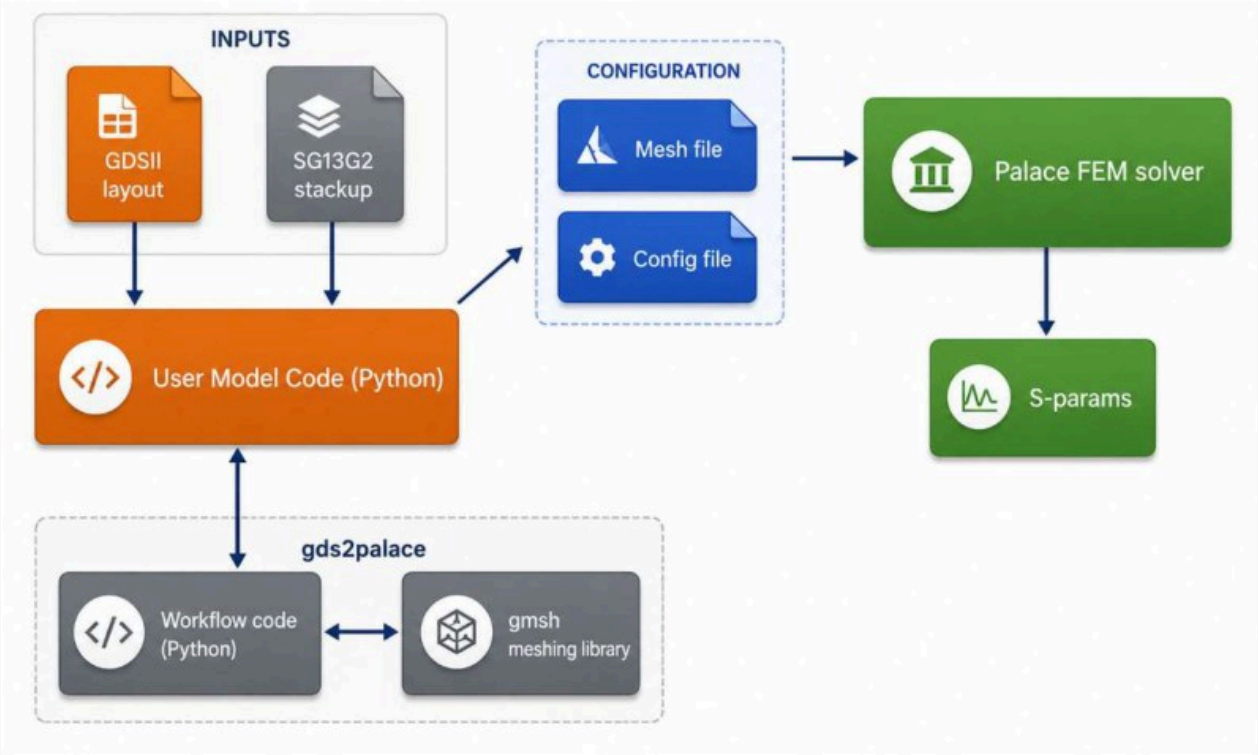


Figure 1: Workflow overview: from GDSII layout, XML stackup and model script to Palace input files

4 Required software and Python modules

The gds2palace Python workflow creates AWS Palace model files (mesh file + config file). To actually simulate these models, you need to have AWS Palace installed. Installing Palace is described below in chapter “Installing AWS Palace”.

4.1 All-in-one installation script for Linux

An installation script for Linux is available, which

- creates the Python venv for gds2palace and its user interface setupEM
- installs gds2palace, setupEM and all dependencies
- installs the AWS Palace solver as an Apptainer container image, and checks that it runs
- installs the scripts to run Palace models from gds2palace (run_palace, combine_snp)

See https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/tree/main/scripts/install_linux

4.2 Manual installation

It is recommended to install gds2palace into a Python virtual environment. In this document, we assume Linux environment and install to a virtual environment `palace` that is located in the user's home directory at `~/venv/palace`

This is how to create that virtual environment:

```
cd ~
python3 -m venv ~/venv/palace
```

4.3 Recommended installation of gds2palace as Python module

The easiest method to install the gds2palace Python workflow is to use the preconfigured package which automatically installs all dependencies (gdspy, gmsh, scikit-rf). We activate the virtual environment and install to there:

```
source ~/venv/palace/bin/activate
pip install --upgrade gds2palace
```

That's all we need to do for installing the gds2palace Python module. If you install gds2palace this way, you only need the gds2palace repository to download the examples and XML stackup files and documentation.

To run a model script using gds2palace, you then activate the venv first:

```
source ~/venv/palace/bin/activate
```

As an alternative, it would also be possible to work with a local copy of the gds2palace directory, as described below. However, this is more complicated and no longer recommended!

4.4 Alternative, no longer recommended: local gds2palace directory

It is possible to use a local directory with gds2palace module code. This directory can be downloaded from the gds2palace github repository and must always be copied to the directory where the *.py simulation model is located. The simulation model must then have a code line to include that local gds2palace directory in the Python path.

In this case, Python module dependencies must be installed manually:

```
source ~/venv/palace/bin/activate
pip install gdspy
pip install gmsh
pip install scikit-rf
```

This virtual environment with installed dependencies can be used to run gds2palace and the scripts that postprocess the Palace simulation results. To activate this Python environment on the command line:

```
source ~/venv/palace/bin/activate
```

4.5 External tools (not Python modules)

Two more tools are needed for parts of the workflow, but are not installed via pip:

- ParaView — to view field-dump output (Palace/Elmer EM) and Elmer thermal result files.

- An MPI implementation — only needed for multi-process Elmer runs (`settings['ELMER_MPI_THREADS']`, see chapter “Installing Elmer FEM”). Use OpenMPI or MPICH on Linux/macOS; on Windows, install Microsoft MPI.

5 Installing AWS Palace

To actually simulate the model created by gds2palace, you need to have AWS Palace installed. Different methods of installing Palace are described at https://awslabs.github.io/palace/stable/install/ but we have compiled the most relevant methods for gds2palace users below.

5.1 Installing the Palace solver using Apptainer

For development of gds2palace, Palace was installed using the Singularity/Apptainer installation method. This was rather simple and straightforward, even with no knowledge about container usage. A step-by-step guide can be found here: building-palace-apptainer.md

A pre-built container image for the last officially released Palace version (0.16) is also available, if you’d rather not build anything yourself. To download it into your current directory:

```
$ apptainer pull palace_016.sif oras://ghcr.io/volkermuehlhaus/palace_016:latest
```

5.2 Installing the Palace solver using spack package manager

Palace can also be created from source with a few simple commands. All tools required by the build process will be downloaded and installed automatically by spack, so you can sit and watch while your system builds the software.
A step-by-step guide for compiling Palace using the **spack package manager for Linux** : building-palace-spack.md

5.3 Running Palace (after installation)

You can use any of the installation methods described on the AWS Palace web site. The gds2palace workflow does not change, it only creates the input files for Palace and does not care how you installed Palace, or on what platform you run the actual Palace simulation from these model files.

To start Palace from setupEM, a wrapper script **run_palace** is used, and this is where you point to your actual installation (even remote copy & remote simulation is possible). This script is available in the scripts folder in gds2palace repository, and needs to be adjusted for your path settings and number of processor cores.

6 Quick tour

This chapter will give a very brief overview of the workflow usage, without going into the details.

To get started, we need a directory that contains:

- GDSII layout file to be simulated
- XML stackup file
- Only if gds2palace is **not** installed as Python module: local copy of gds2palace directory

In addition, we need the simulation model code in Python, which brings the workflow to life. Below are the most important code sections that need to be configured by the user.

6.1 Simulation model: Input files

In this section, the user specifies the GDSII data source and the technology stackup file.

```
# ==================== input files and path settings ======================

gds_filename = "line_simple_viaport.gds"   # geometries
XML_filename = "SG13G2_nosub.xml"          # stackup

# preprocess GDSII for safe handling of cutouts/holes?
preprocess_gds = False
```

6.2 Simulation model: Simulation control

In this section, the user needs to specify frequency range and mesh settings. There are additional optional settings that can be applied, as discussed later in this document.

```
settings['unit']   = 1e-6  # geometry is in microns
settings['margin'] = 50    # distance in microns from GDSII geometry boundary to simulation boundary

settings['fstart']  = 0e9
settings['fstop']   = 100e9
settings['fstep']   = 2.5e9

settings['refined_cellsize'] = 2  # mesh cell size in conductor region
settings['cells_per_wavelength'] = 10   # how many mesh cells per wavelength, must be 10 or more

settings['meshsize_max'] = 70  # microns, override cells_per_wavelength
settings['adaptive_mesh_iterations'] = 0
```

Users who are familiar with the IHP openEMS workflow will notice many similarities, although some implementation details are different.

One important difference is that FEM simulates one frequency after another, so frequency settings have an effect on total simulation time. However, Palace is configured to use an adaptive frequency sweep, which will EM simulate only the minimum required number of frequencies across the band, and then create all other frequency points using interpolation. This adaptive sweep is enabled by default in our workflow.

Another important difference is the use of “refined_cellsize” to create the mesh: the gmsh meshing engine will always create a mesh based on all geometry detail, no matter how small it is, and then apply an additional mesh refinement based on “refined_cellsize” along the metal edges.

6.3 Simulation model: Ports

Additional important user input are the port definitions.

Similar to the IHP openEMS workflow, **ports are created based on polygons from the GDSII file** , located on special layers that are not part of the IHP layer table. Each port needs to use a different source layer in the GDSII file.

The port mapping defines the port number and port impedance, and then maps the port geometry from the special GDSII input layer (usually 201 and above) to actual IHP technology layers. For the vertical ports shown here, we have from_layername and to_layername. For in-plane ports, we would have target_layername instead. Finally, the port direction is required to specify vertical ports or in-plane ports and their direction/polarity.

```
simulation_ports = simulation_setup.all_simulation_ports()
# instead of in-plane port specified with target_layername,
# we here use via port specified with from_layername and to_layername
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=1, voltage=1, port_Z0=50, source_layernum=201,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
```

```
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=2, voltage=1, port_Z0=50, source_layernum=202,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
```

For this Palace workflow, the voltage parameter is not supported yet by the Palace solver. Palace only supports opposite polarity by reversing the port direction. **But here in our workflow, we use this voltage parameter to specify if a port is active: ports with voltage=0 will not be excited.**

It must be noted that Palace behaves differently from other FEM solvers: to get the full S- matrix, we need to run all port excitations, one after another. This only happens if all port voltages are not zero. Only then, the full S-parameter output file can be created.

ATTENTION : If any port excitation is zero, that “zero voltage” port will not be excited and that row in the S-parameter file will be padded with zeros. For example, if we simulate an 8-port circuit and only excite port 1, the S-parameter output will have valid results for S11, S21, … S81 but all other values will be zero. This can be useful to quickly simulate one specific path in the model.

To get the full S-parameter file, all ports must be defined with non-zero voltage, so that all port excitations are simulated, one after another.

PORT SHAPE IN GDSII : The workflow creates lumped ports in Palace, which need to be 2D sheets. This means that for vertical ports (via ports), you should draw a zero-width box in GDSII, resulting in a vertical 2D sheet with no width. If you define a vertical port from a box with finite xy area, the Palace port will be created as a vertical 2D sheet along the center line of that 2D box.

Composite ports (multiple EM ports grouped into one port in the final output file) are not yet supported by this workflow. All ports with non-zero voltage are simulated one after another, resulting in n-port S-parameters.

6.4 Running the model code to create Palace input files

The simulation model file (Python code) can be run on the command line. After reading and processing the input files, a 3D viewer comes up and shows the resulting 3D model. This viewer is the graphical interface of the gmsh meshing library, and provides many options for inspection of the model. At this point, the model is not meshed yet, so that we can see the raw geometries.

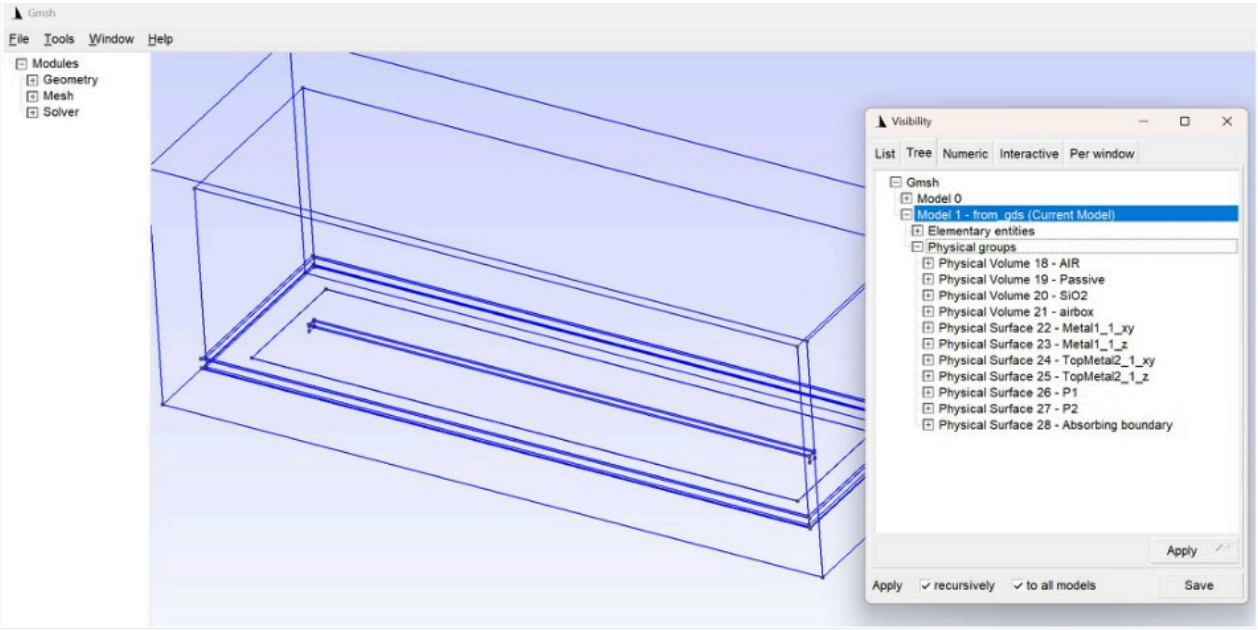


Figure 2: gmsh preview of the raw model geometry, before meshing

To see the structure of the 3D model, you can go to Tools > Visibility.

In the screenshot, you can see that dielectric boxes (SiO2, Passive, AIR etc) have been added around the GDSII polygons, with an offset value in xy direction from the “margins” parameter in the model file. Around that, we have another layer of air on all six sides, using the same margins value.

Metals have been created as surfaces. All polygons on each layer are merged, if possible, and then each of the resulting polygons is created as a separate surface. To be more precise, all faces of each resulting polygon body (top, bottom and side walls) form one surface group, named after the layer and a number, e.g. TopMetal2_1.

You can also see two ports P1 and P2 created as surfaces. The outer simulation boundary is this example is a surface “Absorbing boundary” that is defined as absorbing boundary in the Palace config file.

When we are done with inspecting the model (which is optional, no user action is required!), the gmsh viewer window can be closed to proceed with meshing. After closing the gmsh geometry preview, the simulation model script will mesh these geometries, which can take a while and will show lots of status information on the command line.

When meshing is completed, the gmsh 3D viewer will be displayed again, showing the overall mesh.

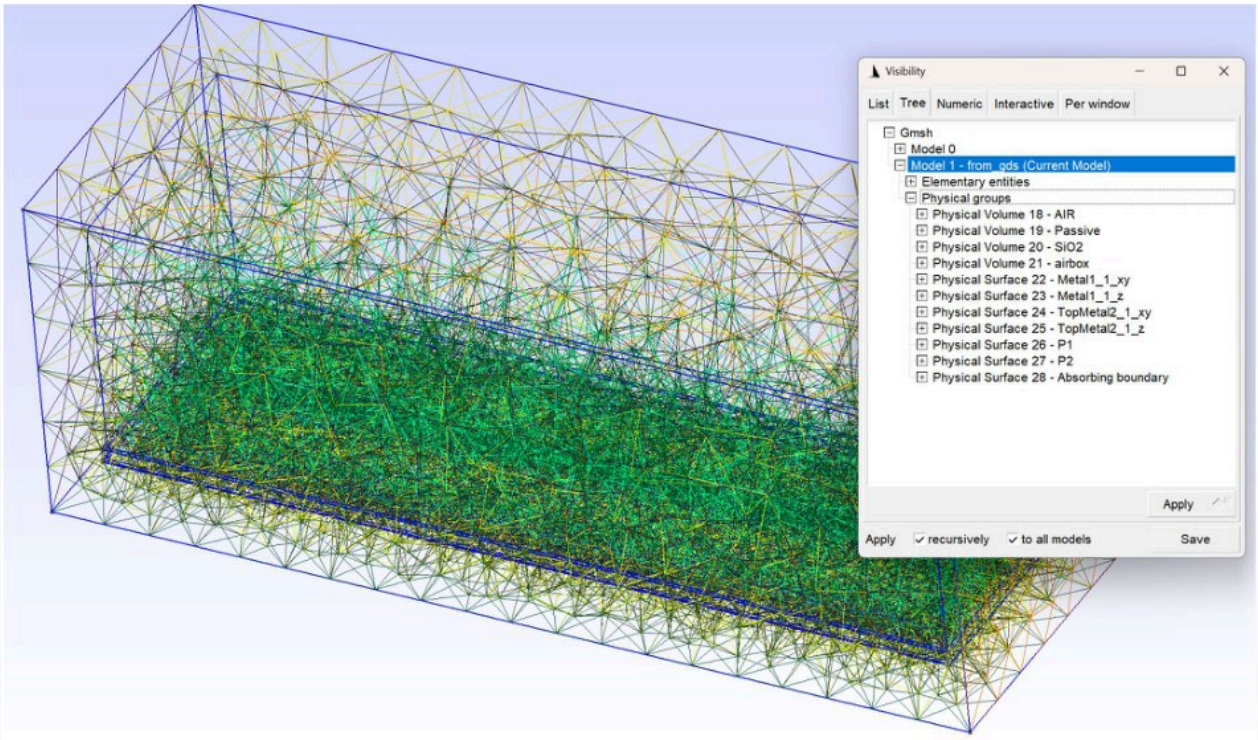


Figure 3: gmsh view of the complete mesh

This is too complex to see anything, but we can now go to Tools > Visibility and select one or more groups to be displayed:

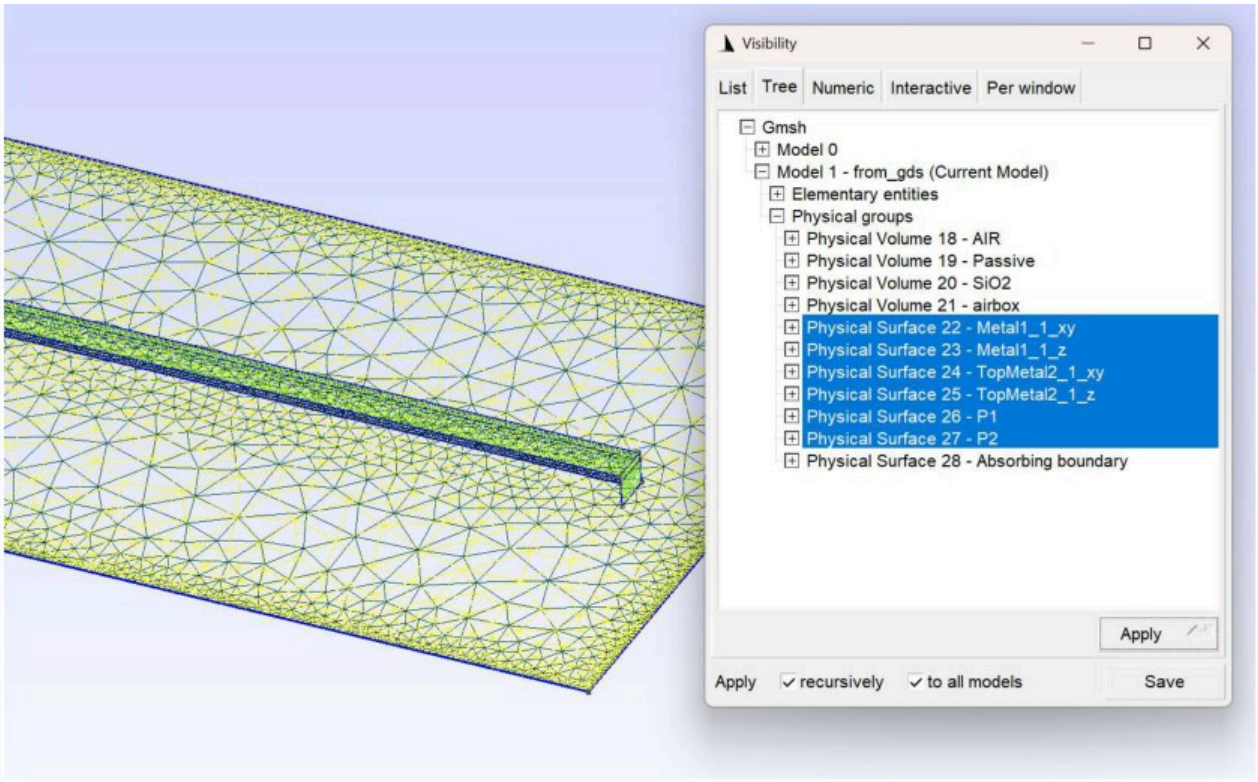


Figure 4: Mesh on the conductor surfaces and ports, selected in Tools > Visibility

This shows the mesh at the conductor surfaces of Metal1, TopMetal1 and the vertical ports. We can also display the meshed oxide, which has a lot of detail because the metal layers reside inside this volume. The metals have been “cut out” from the dielectric layer, and we see the mesh refinement around the conductor edges.

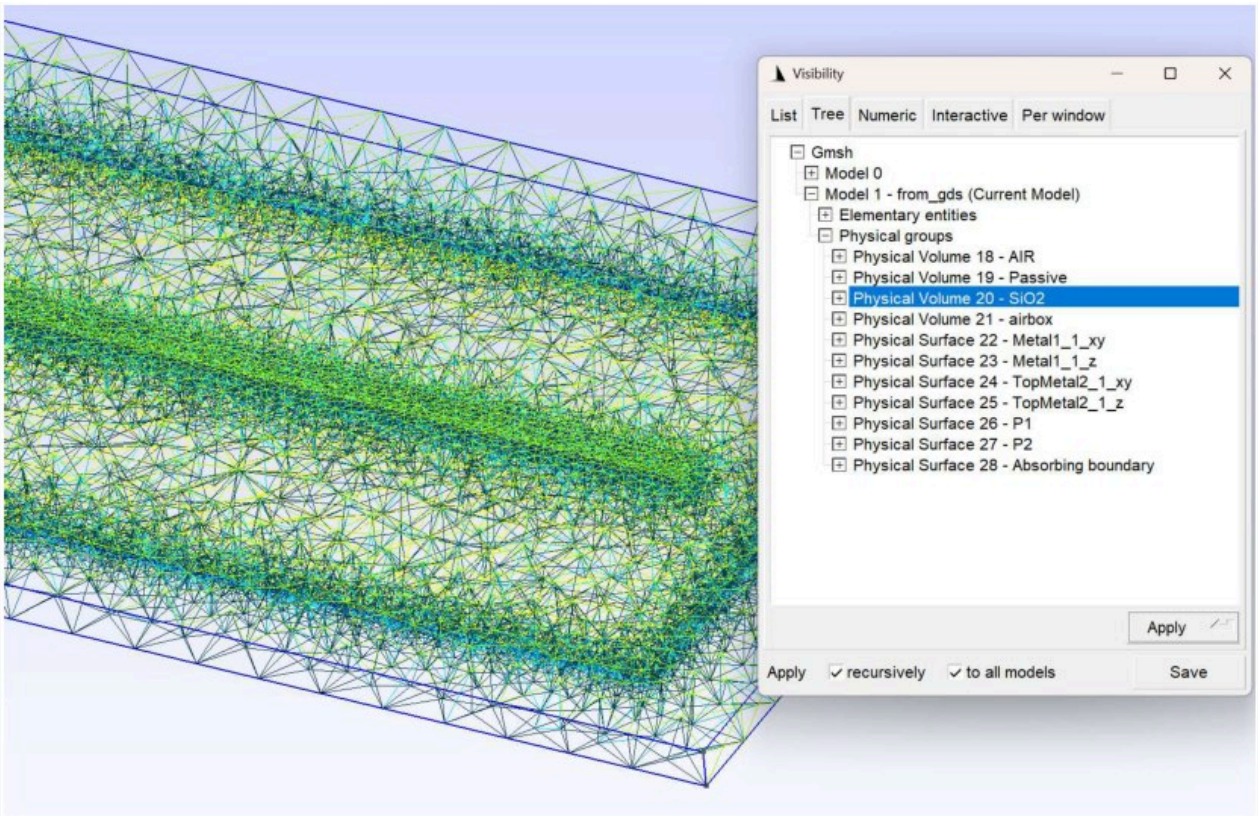


Figure 5: Mesh of the SiO2 volume, with the metals cut out and refined mesh at the conductor edges

When you are done with inspection of the model (which is optional, no user action is required!), the gmsh viewer window can be closed to finish model generation. The workflow code has now created the required file to start Palace: the simulation control file config.json and the mesh file.

```
{
    "Problem": {
        "Type": "Driven",
        "Verbose": 3,
        "Output": "output/palace_line_viaport"
    },
    "Model": {
        "Mesh": "palace_line_viaport.msh",
        "L0": 1e-06,
        "Refinement": {}
    },
    "Solver": {
        "Driven": {
            "MinFreq": 0.1,
            "MaxFreq": 100.0,
            "FreqStep": 2.5,
            "SaveStep": 0,
            "AdaptiveTol": 0.01
        },
        "Linear": {
            "Type": "Default",
            "KSPType": "GMRES",
            "Tol": 1e-06,
            "MaxIts": 400,
            ...
```

6.5 Running Palace FEM simulation from our input files

To simplify running the solver, in addition to mesh file and config file, a script file was also created named *run_sim*.

```
#!/bin/bash
run_palace config.json
combine_snp
```

This starts a script *run_palace* where you can define in detail how to run Palace. It is recommended that you place this run_palace script in your PATH, configured for the actual machine where you want to run Palace.

For workflow development and test, Palace was installed into an apptainer container, as documented in the Palace documentation in chapter "**Build using Singularity/Apptainer**".

The script *run_palace* was then configured to start Palace inside the palace_014.sif container, which is located in the user's home directory, and passes one command line argument (the config.json file). The additional parameter *-np 16* tells palace to run using 16 threads. This will partition the simulation domain into 8 pieces, each running in a separate process, and the combine results into one output directory.

```
#!/bin/bash
apptainer exec ~/palace_014.sif palace -np 16 $1
```

Running Palace is not very spectacular, the simulation progress is shown in the terminal window.

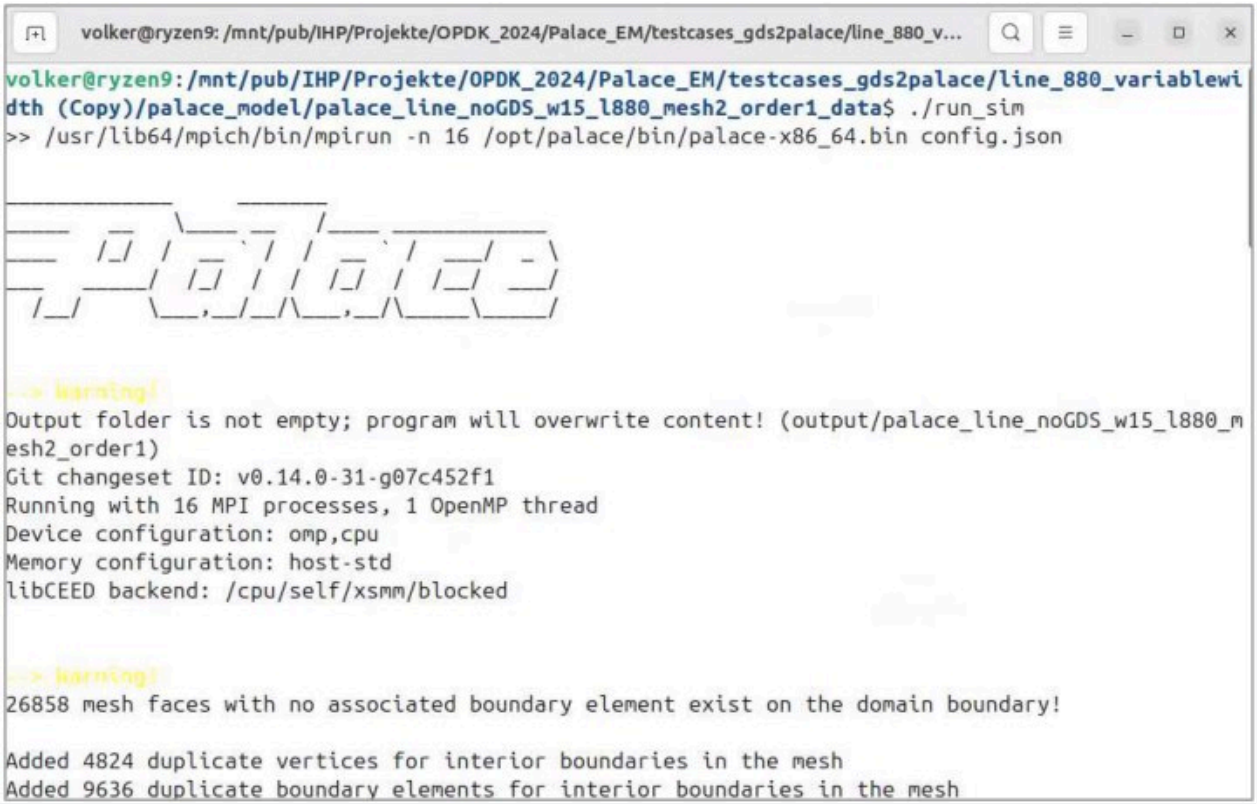


Figure 6: Palace simulation progress in the terminal window

When *run_palace* is finished, Palace output files are created in the *output* directory below the simulation model directory. S-parameters are in *.csv file format, which we need to convert now.



Figure 7: Palace output files

To do that conversion, the run_sim script where we started simulation with “run_palace” executes another command *combine_snp*.

combine_snp is a script that searches for Palace s-parameter files (port-S.csv) and converts them to Touchstone file format. If the model was simulated for selected port excitations only, missing rows in the S-parameter file will be padded with zeros.

It should be noted that Palace frequency domain simulation cannot go down to 0 Hz, and the workflow script will replace any 0 Hz start frequency by a low value like 1 GHz.

If simulation results include such low frequency data, combine_snp will create an additional S-Parameter file with suffix "_dc", with a DC value extrapolated from the EM simulated data.[1] Always check that DC extrapolated dataset carefully before use!



Figure 8: Output files after combine_snp: Touchstone file and DC-extrapolated Touchstone file

This ends our quick tour of using the gds2palace workflow. There are many additional settings available, which will be described in detail in the next chapters.

7 Simulation model file in detail

7.1 Input files

Typical simulation models require two input files: GDSII geometries and XML stackup.

```
# ====================== input files and path settings =======================

gds_filename = "BM_Ardavan_Rahimian_with_ports.gds"   # geometries
XML_filename = "SG13G2_nosub.xml"          # stackup
```

Two settings are available to control processing of the GDSII geometries:

```
# preprocess GDSII for safe handling of cutouts/holes?
preprocess_gds = False

# merge via polygons with distance less than .. microns, set to 0 to disable via merging.
merge_polygon_size = 0
```

preprocess_gds is no longer needed: layouts with cutouts (holes) or other self-intersecting polygon boundaries are now handled automatically. If an older model script still sets it to True, the GDSII reader ignores it and prints a note. Earlier versions needed it, otherwise meshing of layouts with holes failed with errors like "Exception: Curve loop is not closed".

merge_polygon_size applies to layers which are declared as *Type="via"* in the XML stackup file.

[1] combine_snp is a Python script created for this workflow. DC extrapolation in this script is powered by Python library scikit-rf.

When enabled with a non-zero value, the code will merge polygons on these layers with the given distance. This is how it works: in the GDSII reader, polygons on this layer will be oversized by half the given value, then all overlapping polygons on that layer will be merged, then undersized by half the given value.

For via arrays that are oriented along the xy-axis, this will give the resulting bounding box if the maximum via spacing is no larger than the given value.

Merging turns the gaps between the vias into solid via material, so the merged polygon contains more via metal than the real via array. By default, it still gets the full via conductivity from the XML stackup, which overestimates the conductivity of the via array: in the IHP stackup files, via conductivity is calculated from the resistance per via in the process specification, so it describes a single via, not a via array. With settings['fill_factor_correction'] = True (see settings), the conductivity of each merged via polygon is multiplied by its fill factor (original via area / merged polygon area).

This correction only knows about merging done here, by merge_polygon_size. If the via arrays were already merged before, e.g. with gds_prepare_for_EM, they arrive as solid polygons with fill factor 1.0, and their conductivity is not corrected.

7.2 settings

settings for meshing and simulation control are implemented as a Python dictionary, which is passed to the simulation_setup.create_palace function. Some settings are required, other are optional with a meaningful default value.

These settings are always required:

settings['unit']: Unit of values in mesh, typically 1E-6
settings['margin']: Oversize of dielectrics from bounding box of drawing in xy plane
settings['fstart']: Start frequency in Hz
settings['fstop']: Stop frequency in Hz
settings['fstep']: Frequency step in Hz for output, not all of them need to be EM simulated due to adaptive frequency sweep
settings['refined_cellsize']: Target mesh size at polygon edges

All other settings are optional, grouped here by what they control.

7.2.1 Frequencies

Discrete frequencies can be used in addition to fstart/fstop and fstep, or instead of fstart/fstop and fstep:

settings['fpoint']: Discrete frequency/frequencies, values enclosed in [ ]. Example: settings['fpoint']=[10e9, 15e9]
settings['fdump']: Same as fpoint, but Palace is configured to write a field dump for Paraview at this frequency/these frequencies. Example: settings['fdump']=[10e9]
settings['adaptive_sweep']: Enable adaptive frequency sweep, default is True

7.2.2 Mesh size and accuracy

settings['cells_per_wavelength']: Calculated at highest frequency,value must be 10 or more, default is 10
settings['meshsize_max']: Maximum mesh size limit, in addition to cells, default is 70
settings['refined_cellsize_override']: Optional per-layer override of refined_cellsize,example: settings['refined_cellsize_override']=[['Metal3',10],['Metal2',2]]
settings['substrate_refinement']: Extra mesh refinement into substrate, usually not required, default is False
settings['order']: Order of basis function for FEM solver (order 2 is more accurate,order 1 is only for quick & dirty results). Default is 2

7.2.3 Adaptive mesh refinement

settings['adaptive_mesh_iterations']: Iterations for adaptive mesh refinement, often not required when using fine initial mesh, default is 0
settings['adaptive_mesh_conformal']: Use conformal AMR refinement instead of Palace's default noncon-

formal (hanging-node) refinement, default is False
settings['amr_tol']: Target relative error for adaptive mesh refinement: refinement stops once the error estimate is below this value. Only used with adaptive_mesh_iterations > 0, default is 0.01
settings['amr_max_dof']: Maximum number of unknowns for adaptive mesh refinement: refinement stops once the model reaches this size. Only used with adaptive_mesh_iterations > 0, default is 2e6
settings['save_adaptive_mesh']: Save mesh file from adaptive iteration for possible re-use, default is False

7.2.4 Linear solver (Palace)

settings['complex_coarse_solve']: Let Palace's sparse direct coarse solver factorize the full complex system, instead of only its real part (Palace's own default). **Default is True, for every model**, whether metals are meshed as surface impedance or filled. The real-part-only preconditioner ignores absorbing boundaries, port resistances and losses. Some models then need far more iterations or don't converge at all, and a frequency that doesn't converge gets **wrong S-parameters**, not just slightly inaccurate ones. Which models are affected can't be predicted from the settings: it happened at high frequency (a D-band grounded CPW line failed at 170 GHz) and at a single low frequency (an inductor failed at 2.4 GHz and gave 1.1 nH instead of 2.5 nH, while the same model converged at 0.1 and 10 GHz). With True, both converged in 1-31 iterations and ran 3.5-6x faster. Models that already converge without it get identical results, in about the same or less time. The cost is peak RAM, which depends on the model: 1.0-1.6x at order 2 and 1.5-1.9x at order 1 in our test cases. Set it to False only if a model doesn't fit into memory, and then check the Palace output for `did NOT converge`
settings['solver_maxits']: Iteration limit of Palace's linear solver (GMRES) per frequency. If it is reached, Palace only logs "solver did NOT converge" and carries on, and the S-parameters at that frequency are unreliable. Default is 400
settings['solver_tol']: Relative residual tolerance of Palace's linear solver. Default is 1e-6

7.2.5 Simulation boundary

settings['boundary']: List with 6 values for boundary at xmin,xmax,ymin,ymax,zmin,zmax. Values can be ABC/PML, PEC or PMC. Default: ['ABC','ABC','ABC','ABC','ABC','ABC']
settings['air_around']: Other spacing of air around dielectrics, default is same as margin. Can be a single value or a list of 6 values [air_xmin, air_xmax, air_ymin, air_ymax, air_zmin, air_zmax]. A value of 0 is allowed on any side, placing the simulation boundary flush with the dielectric/metal stack on that side (e.g. a backside ground plane serving directly as the PEC/ABC boundary, with no wasted air layer below it).

7.2.6 Conductor modelling

settings['conductor_model']: The conductor model in one choice, named as in HFSS: `'surface'` (default) puts a surface impedance on all conductor faces, the fastest model; `'solve_inside'` meshes the conductors as volumes (= `filled_metals = True`); `'solve_inside_skin'` adds thin layers along their faces for the skin effect (= `filled_metals = True` and `skin_layers = True`). For Q and loss, use `'solve_inside'` while designing and `'solve_inside_skin'` for final numbers; `'surface'` is fine for inductance, coupling, match and balance. Errors, run times and what they mean for typical designs: see "Choosing the conductor model" in chapter "Conductor loss modelling". `filled_metals` and `skin_layers` below are the expert settings behind it; when given, they override the preset. Default is 'surface'

settings['filled_metals']: Expert setting, set by `conductor_model`. Model conductors as solid volumes with bulk conductivity (volume mesh) instead of a surface impedance on the conductor surfaces. This gives the right conductor loss at low frequency, where the skin depth is no longer small compared to the conductor cross section. At high frequency, where the skin depth is a fraction of the cell size, the loss is a few percent too high at a 2 µm mesh (see `skin_layers`). Needs more RAM and simulation time than the surface model. Tested with Palace, Elmer thermal models always use solid conductor volumes. Default is False

`settings['filled_metals'] = 'auto'` meshes only the conductor layers thinner than `refined_cellsize` as volumes and keeps the others as surface impedance. A thin layer is meshed only about one cell thick anyway, so a volume costs little, and the surface model overestimates the loss of thin layers that carry current on one face only, up to twice (see "Palace's External option"). With `refined_cellsize` 2 in SG13G2, Metal1-Metal5, MIM and Activ are filled, TopMetal1 and TopMetal2 stay surface impedance. The choice does not depend on the frequency range, so an over-specified highest frequency does not change it. MIM capacitor `palace_rfcmim.py`:

series resistance at low frequency (10 MHz) 0.85 Ω with 'auto', 0.87 Ω with all metals filled, 1.88 Ω with surface impedance, at about the run time of the surface model. `palace_line_viaport.py` (TopMetal2 line over a Metal1 ground plane): line resistance 3-11 % too low with 'auto' (the surface model error of TopMetal2), instead of 16-41 % too high with surface impedance. Use 'auto' with care: coils on TopMetal1/TopMetal2 keep the Q error of the surface model, and on a MIM-loaded balun of the mesh convergence studies, where the two errors of the surface model (thick metals too little loss, MIM plates too much) cancelled, 'auto' removed only one of them and was 0.24 dB too optimistic.

settings['skin_layers']: Expert setting, set by `conductor_model = 'solve_inside_skin'`. Only with `filled_metals`. Splits each conductor into thin layers along all its faces (top, bottom and side walls) plus a core, so that the mesh has thin cells next to the conductor surface, where the skin effect puts the current, while the cell size along the surface stays at `refined_cellsize`. Without it, filled metals need a small `refined_cellsize` everywhere to be accurate at high frequency. Values:

- `True`: all conductor layers that are at least 4 skin depths thick at the highest simulation frequency (in SG13G2: TopMetal2 from about 15 GHz, TopMetal1 from about 35 GHz, Metal1-Metal5 not below about 1 THz), depths of about 0.4 and 1 skin depth, from the layer's conductivity
- a list of depths in drawing units, e.g. `[0.1, 0.25]`: these depths for the same layers as `True`
- a dictionary for selected layers, e.g. `{'TopMetal2': True, 'TopMetal1': [0.1, 0.25]}`: recommended, because the skin layers are only needed where the conductor loss matters, and each layer adds mesh cells

A depth is only used if a core of at least 1/3 of the layer thickness remains; the model script prints the depths used per layer. PEC layers and vias are not split. With `True` or a list, layers thinner than 4 skin depths are not split either: they gain little, while their very thin skin layers cost much solver effort. For the 0.42 µm Metal1 ground plane of `palace_line_viaport.py`, skin layers changed the line resistance by about 1 % and made the simulation 2.6 times slower; in the D-band balun of the mesh convergence studies, skin layers in Metal3-Metal5 left the result unchanged and made it 1.7 times slower. Neither are layers with a conductivity above 1e8 S/m: no real metal conducts better than silver (6.3e7 S/m), so such a material stands in for a perfect conductor, e.g. `LOWLOSS` (1e10 S/m) for SUBGND in older stackups, and its skin layers would be a few nanometres thin. Tested on TopMetal2 inductors and lines up to 150 GHz with `refined_cellsize` 2, see chapter "Conductor loss modelling". Default is False

settings['fill_factor_correction']: Only has an effect with merge_polygon_size > 0. Via array merging fills the gaps between vias with via material, so each merged via polygon is scaled down to its real via conductivity: its conductivity is multiplied by its fill factor (original via area / merged polygon area). Merged vias on the same layer are grouped into separate materials, starting a new group when the fill factor is more than 20% below the group's highest value, and each group uses its mean fill factor. Vias with the reserved PEC material are grouped but not scaled. Applies to Palace and Elmer EM (electrical conductivity) and to Elmer thermal (heat conductivity, including temperature table values). Default is False

7.2.7 Script control and output files

settings['no_gui']: Run script without showing gmsh user interface, useful for automated processing, default is False

settings['no_preview']: don't show unmeshed geometry, immediately show meshed model, default is False

settings['preview_only']: Only show the unmeshed geometry in gmsh, then stop without generating a mesh, default is False

settings['save_gmsh_unrolled']: Also save gmsh geometry file without meshing, for later inspection, default is False

settings['config_suffix']: Text appended to the names of the generated config.json and port_information.json files, e.g. '_fine' gives config_fine.json, default is '' (no suffix)

7.2.8 Model data set by the script

The model script also passes the stackup and layout data in the settings dictionary. These are not options: the script sets them after reading the XML stackup and the GDSII file.

settings['materials_list'], settings['dielectrics_list'], settings['metals_list']: Stackup data, as returned by stackup_reader.read_substrate()

settings['allpolygons']: Layout polygons, as returned by gds_reader.read_gds()
settings['simulation_ports']: Port definitions, created with simulation_setup.all_simulation_ports()
settings['sim_path']: Output directory for the generated model files
settings['model_basename']: Base name of the generated model files, e.g. the mesh file model_basename.msh

7.3 Port configuration

Similar to the IHP openEMS workflow, **ports are created based on polygons from the GDSII file** , located on special layers that are not part of the IHP layer table. Each port needs to use a different source layer in the GDSII file.

The port mapping defines the port number and port impedance, and then maps the port geometry from the special GDSII input layer (usually 201 and above) to actual IHP technology layers. For the vertical ports shown here, we have from_layername and to_layername. For in-plane ports, we would have target_layername instead. Finally, the port direction is required to specify vertical ports or in-plane ports and their direction/ polarity.

```
simulation_ports = simulation_setup.all_simulation_ports()
# instead of in-plane port specified with target_layername,
# we here use via port specified with from_layername and to_layername
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=1, voltage=1, port_Z0=50, source_layernum=201,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=2, voltage=1, port_Z0=50, source_layernum=202,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
```

For this Palace workflow, the voltage parameter is not supported yet by the Palace solver. Palace only supports opposite polarity by reversing the port direction. **But here in our workflow, we use this voltage parameter to specify if a port is active: ports with voltage=0 will not be excited.**

It must be noted that Palace behaves differently from other FEM solvers: to get the full S- matrix, we need to run all port excitations, one after another. This only happens if all port voltages are not zero. Only then, the full S-parameter output file can be created.

ATTENTION : If any port excitation is zero, that "zero voltage" port will not be excited and that row in the S-parameter file will be padded with zeros. For example, if we simulate an 8-port circuit and only excite port 1, the S-parameter output will have valid results for S11, S21, … S81 but all other values will be zero. This can be useful to quickly simulate one specific path in the model.

To get the full S-parameter file, all ports must be defined with non-zero voltage, so that all port excitations are simulated, one after another.

PORT SHAPE IN GDSII : The workflow creates lumped ports in Palace, which need to be 2D sheets. This means that for vertical ports (via ports), you should draw a zero-width box in GDSII, resulting in a vertical 2D sheet with no width. If you define a vertical port from a box with finite xy area, the Palace port will be created as a vertical 2D sheet along the center line of that 2D box.

Composite ports (multiple EM ports grouped into one port in the final output file) are not yet supported by this workflow. All ports with non-zero voltage are simulated one after another, resulting in n-port S-parameters.

Port parasitics in results, lumped port vs. wave port: This workflow creates lumped ports, which introduce some physical length into the simulation model, leading to extra path length (? inductance) in results. Other 3D volume meshing EM tools behave the same for lumped ports. If port size is not small compared to the device under test, one possible solution is to estimate these parasitics and remove them in postprocessing (or result use) by cascading negative inductance or negative length at each port.

7.4 Filenames and flow control

If you want to have additional control over the names of files and directories that are created by the workflow, have a look at this section at the beginning of the simulation model file:

```
# get path for this simulation file
script_path = utilities.get_script_path(__file__)

# use script filename as model basename
model_basename = utilities.get_basename(__file__)

# set and create directory for simulation output
sim_path = utilities.create_sim_path (script_path,model_basename)
print('Simulation data directory: ', sim_path)

# change path to models script path
modelDir = os.path.dirname(os.path.abspath(__file__))
os.chdir(modelDir)
```

For example, if you want the customize the model name, to reflect settings used for meshing, you can define these values as variables and append that to the model_basename variable.

```
mesh  = 5  # define here, so that we can add this information to output filenames
order = 3  # define here, so that we can add this information to output filenames

XML_filename = "SG13G2_nosub.xml"          # stackup

# preprocess GDSII for safe handling of cutouts/holes?
preprocess_gds = False

# merge via polygons with distance less than .. microns, set to 0 to disable via merging.
merge_polygon_size = 0

# get path for this simulation file
script_path = utilities.get_script_path(__file__)

# use script filename as model basename, with parameters for suffix
model_basename = utilities.get_basename(__file__) + '_mesh' + str(mesh) + '_order' + str(order)

# set and create directory for simulation output
sim_path = utilities.create_sim_path (script_path,model_basename)
print('Simulation data directory: ', sim_path)

# change path to models script path
modelDir = os.path.dirname(os.path.abspath(__file__))
os.chdir(modelDir)

# ======================= simulation settings ================================

settings = {}

settings['unit']   = 1e-6  # geometry is in microns
settings['margin'] = 50    # distance in microns from GDSII geometry boundary to simulation boundary

settings['fstart']  = 0e9
settings['fstop']   = 100e9
settings['fstep']   = 2.5e9

settings['refined_cellsize'] = mesh  # mesh cell size in conductor region
settings['cells_per_wavelength'] = 10   # how many mesh cells per wavelength, must be 10 or more

settings['meshsize_max'] = 70  # microns, override cells_per_wavelength
settings['adaptive_mesh_iterations'] = 0
settings['order'] = order
```

If you want the script to create the Palace files without showing the mesh in gmsh 3D viewer, this is also possible.

You can use the optional **settings['no_gui']** and either set this to *True* (always run without graphical user interface) or you can check for an optional command line parameter passed to the simulation model script.

```
settings['no_gui'] = ('nogui' in sys.argv)  # check if nogui specified on command line
```

In this case, running the model as

```
python mymodel.py
```

would run with 3D mesh viewer, or running the model as

```
python mymodel.py nogui
```

would run without the 3D viewer. Maybe this gives some idea what customization is possible.

8 Examples

This chapter gives examples on configuration settings, especially mesh control, and their results.

8.1 Single microstrip line

The first testcase for meshing is a microstrip line with TopMetal2 over Metal1 ground. Width is 15 µm to achieve 50 Ohm line impedance, length is 880µm.
This model is simulated with mesh configurations that vary 2 mesh parameters:

- **'refined_cellsize'** defines the mesh resolution along the conductor edges
- **'order** ' defines the order of basis functions, i.e. the degrees of freedom for field variation within one cell

By default, we want to simulate with an order=2, which is a good trade-off between accuracy and speed. However, in this simple example, we will check how these two values can be configured for accurate results, and what this does to simulation time.

The basic idea is this: if we have a higher order of basis functions, the fields can be modelled more accurately within each mesh cell, so we need less mesh cells ? larger value of refined_cellsize.

If we use first order basis functions, we need higher mesh density to accurately model the field distribution. Mesh on conductors at **'refined_cellsize'** = 5:

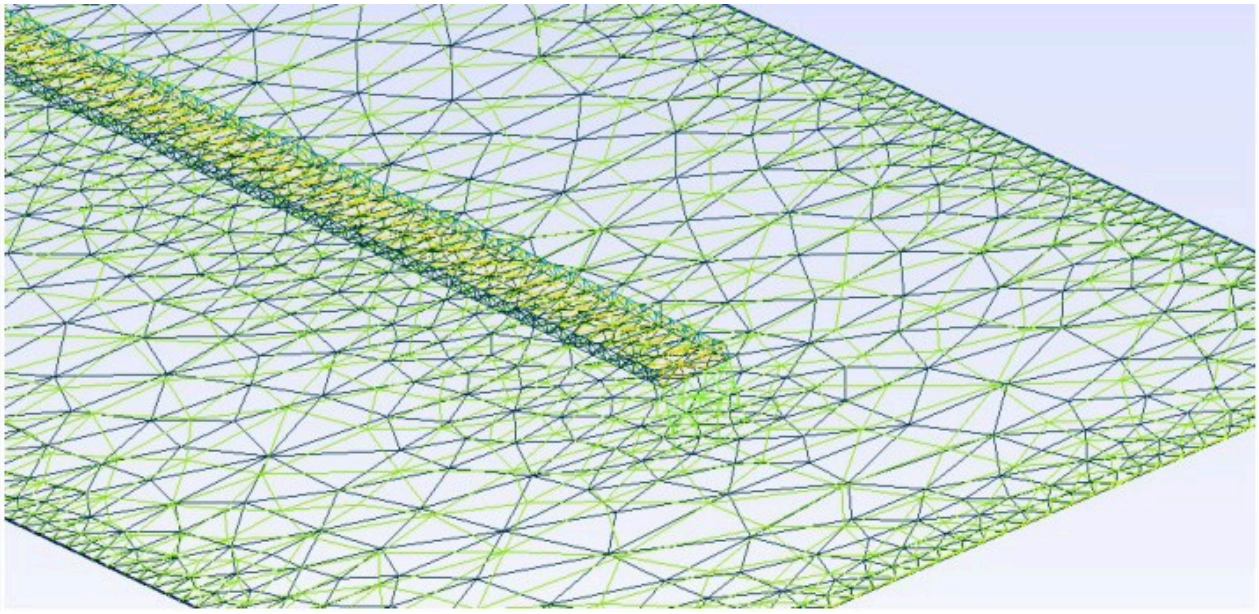


Figure 9: Mesh on the conductors with refined_cellsize = 5

Mesh on conductors at **'refined_cellsize'** = 2:

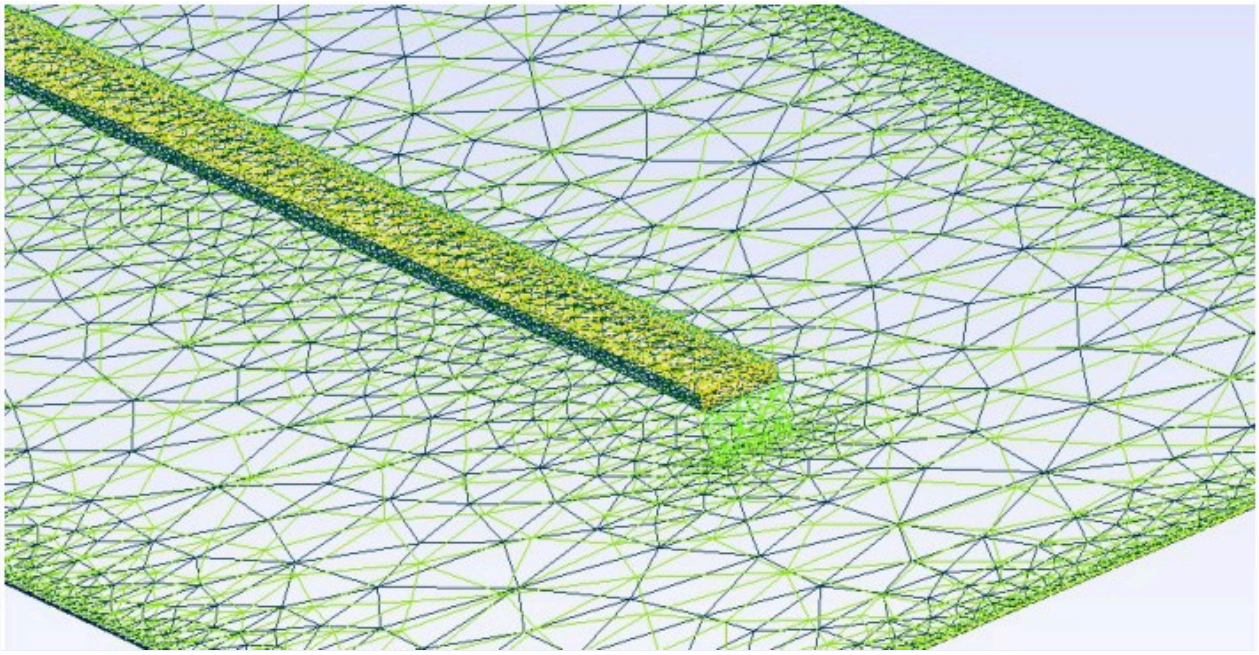


Figure 10: Mesh on the conductors with refined_cellsize = 2

These two mesh densities are now simulated with different order of basis function.

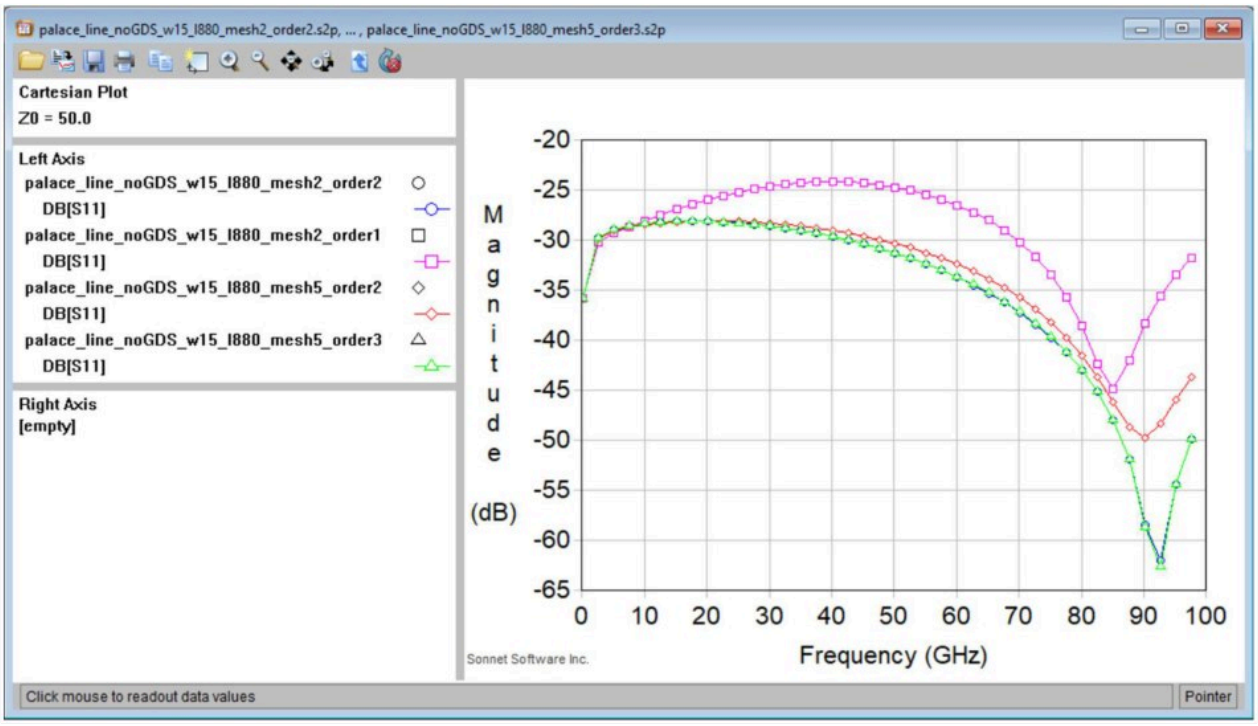


Figure 11: Microstrip line: S11 for different mesh size and basis function order

The clear outlier in this return loss plot is the pink curve, simulated with mesh order = 1 at 2µm refined cellsize. This result is different from the other curves obtained with higher mesh order. Our default simulation would be the blue curve, obtained with refined_cellsize = 2µm and order = 2.
Going to a larger value of refined_cellsize = 5µm (less mesh density at the edges) at order = 2 gives slightly different results, shown by the red curve.

If we combine the larger refined_cellsize = 5µm with higher order = 3, to give each cell more degrees of freedom, we get similar results as our baseline (default) simulation: the green and blue curve are visually identical for S11. For S21 transmission, we get a similar result: these two are almost identical.

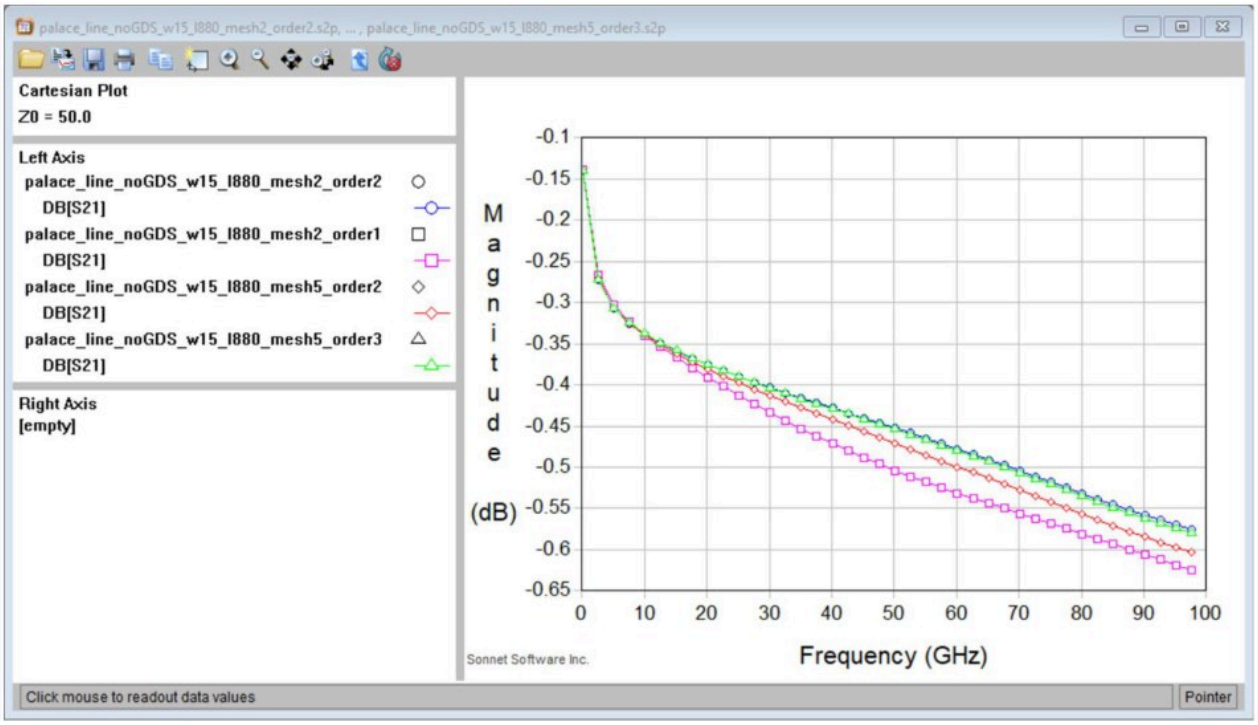


Figure 12: Microstrip line: S21 magnitude for different mesh size and basis function order

For phase of S21 we also have similar results: the order = 1 result is a clear outlier, but all other results are very close.

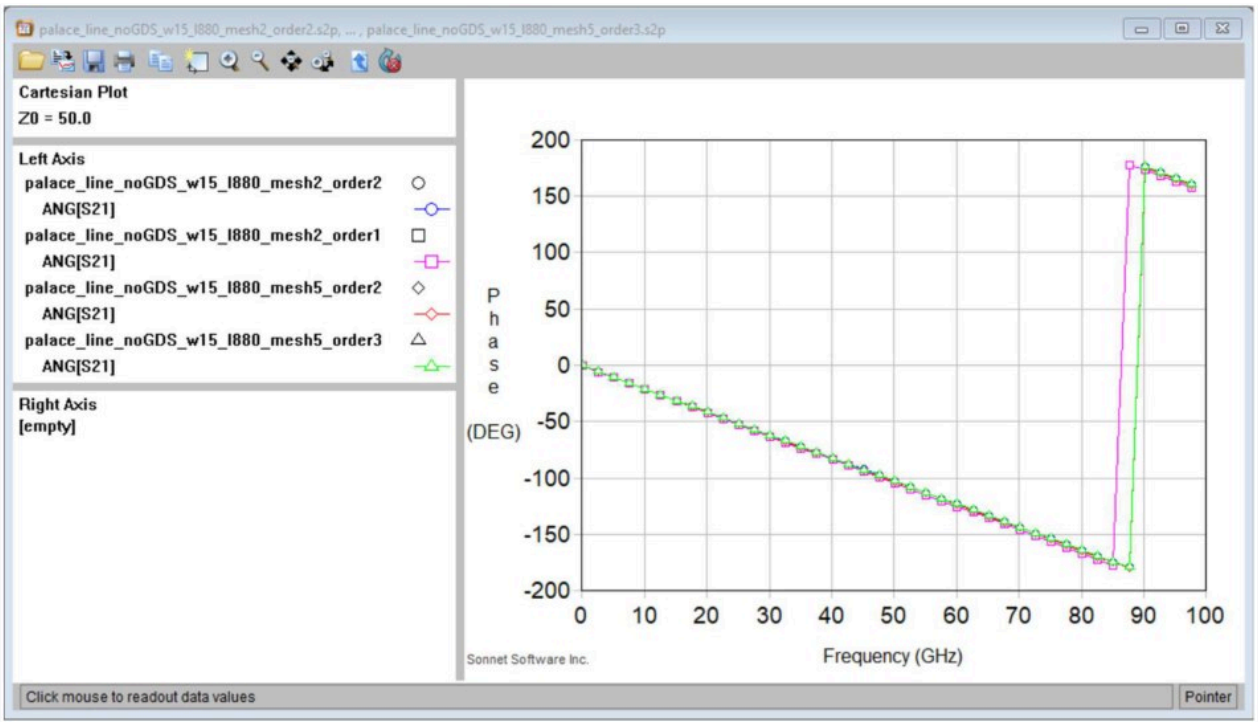


Figure 13: Microstrip line: S21 phase for different mesh size and basis function order

Conclusion: refined_cellsize=2 at order=2 is a good starting point, and for this example we don't need to go any further. Decreasing the mesh resolution to refined_cellsize=5 at order=2 is possible if we can accept some small change in results. Let's look at simulation times for these 4 different meshing approaches:

Simulation time total, Degrees of freedom (from log)
refined_cellsize=2, order=2: 733s, DOF 1640126
refined_cellsize=2, order=1: 64s, DOF 321999
refined_cellsize=5, order=2: 282s, DOF 699658
refined_cellsize=5, order=3: 1967s, DOF 1982661

All simulations were done using adaptive frequency sweep and required 10 discrete frequencies to be EM simulated. Total time in the table also includes interpolation to the specified sweep range. It can be seen that refined_cellsize=2, order=2 is the best choice for accurate data, we get results that are visually identical to refined_cellsize=5, order=3 at less simulation time (733 s total vs. 1967 s total).

In the next chapter, we will go from that baseline setting, and investigate the use of adaptive mesh refinement.

8.2 Balun 140-170 GHz

This testcase is a balun from the phase shifter project by Rupok Das, published for the OPDK July 2025 tapeout:

- https://github.com/IHP-GmbH/TO_July2025/tree/main/FMD_QNC_D_Band_Phase_Shifter



Figure 14: Balun 140-170 GHz: layout

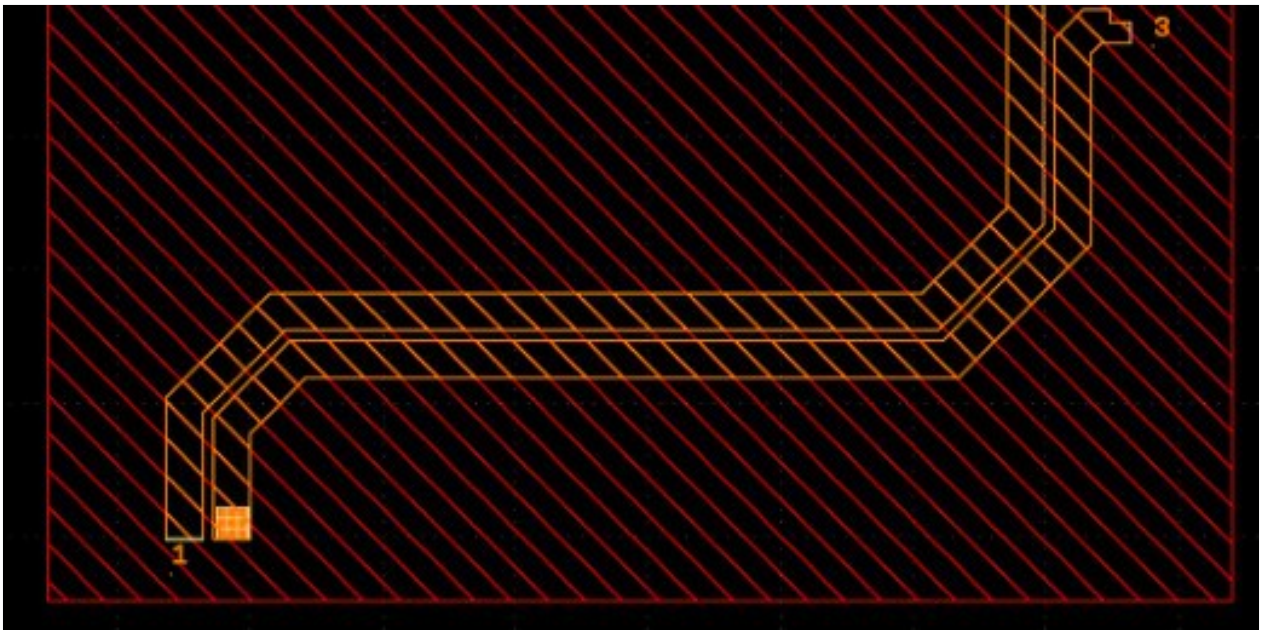


Figure 15: Balun 140-170 GHz: layout with port numbers

Line width is 7µm with TopMetal2 over Metal3 ground, gap width is 2µm.

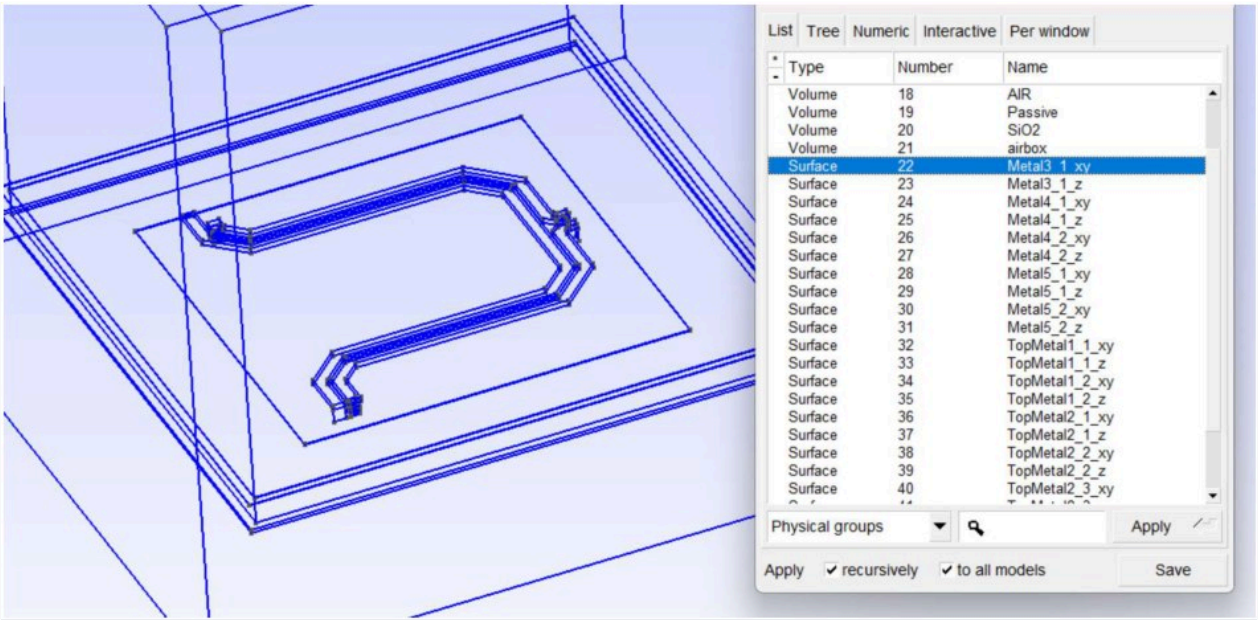


Figure 16: Balun 140-170 GHz: model geometry in gmsh

The critical dimension in this design is the 2µm wide gap between the coupler traces, the solver needs to model the field distribution in that gap with sufficient accuracy.

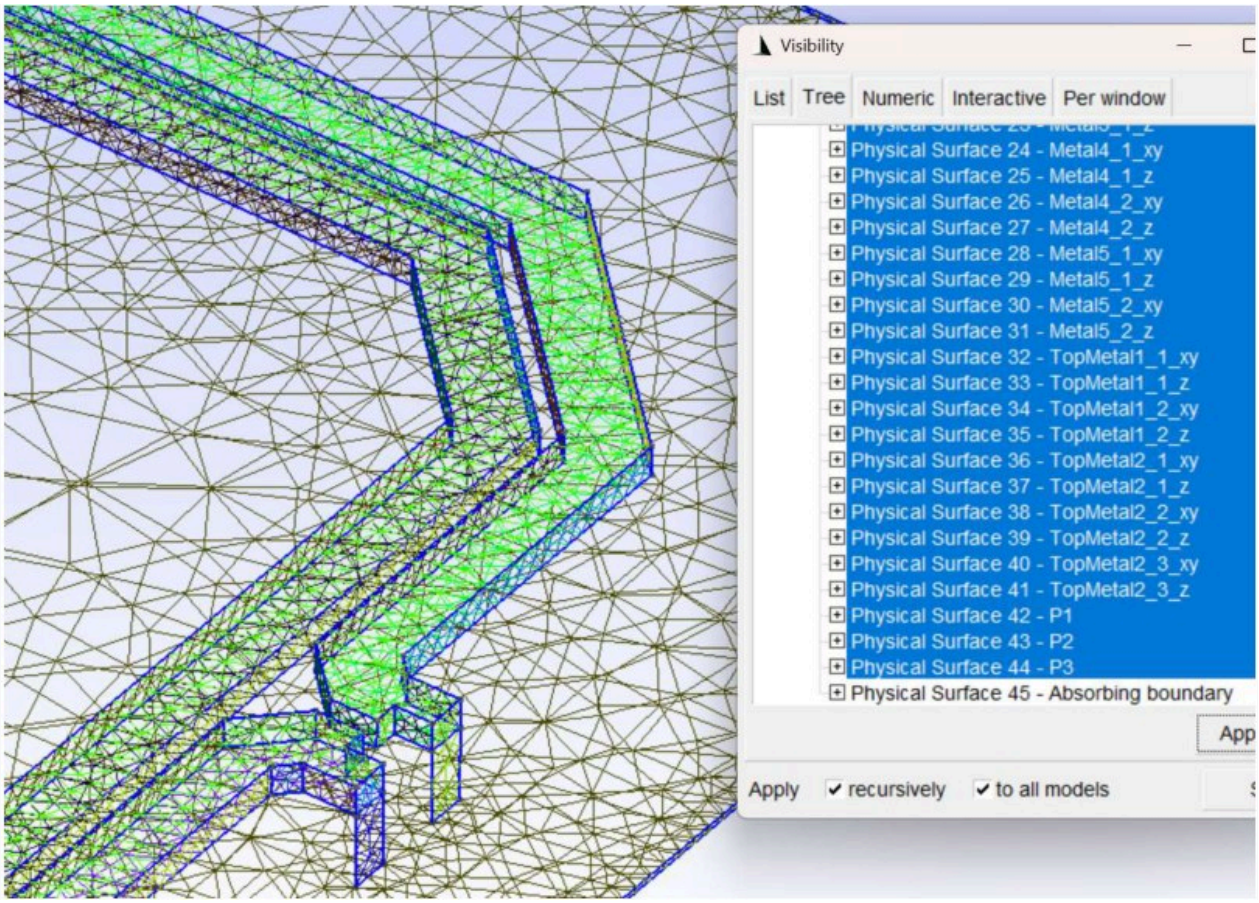


Figure 17: Balun 140-170 GHz: mesh on the conductor surfaces

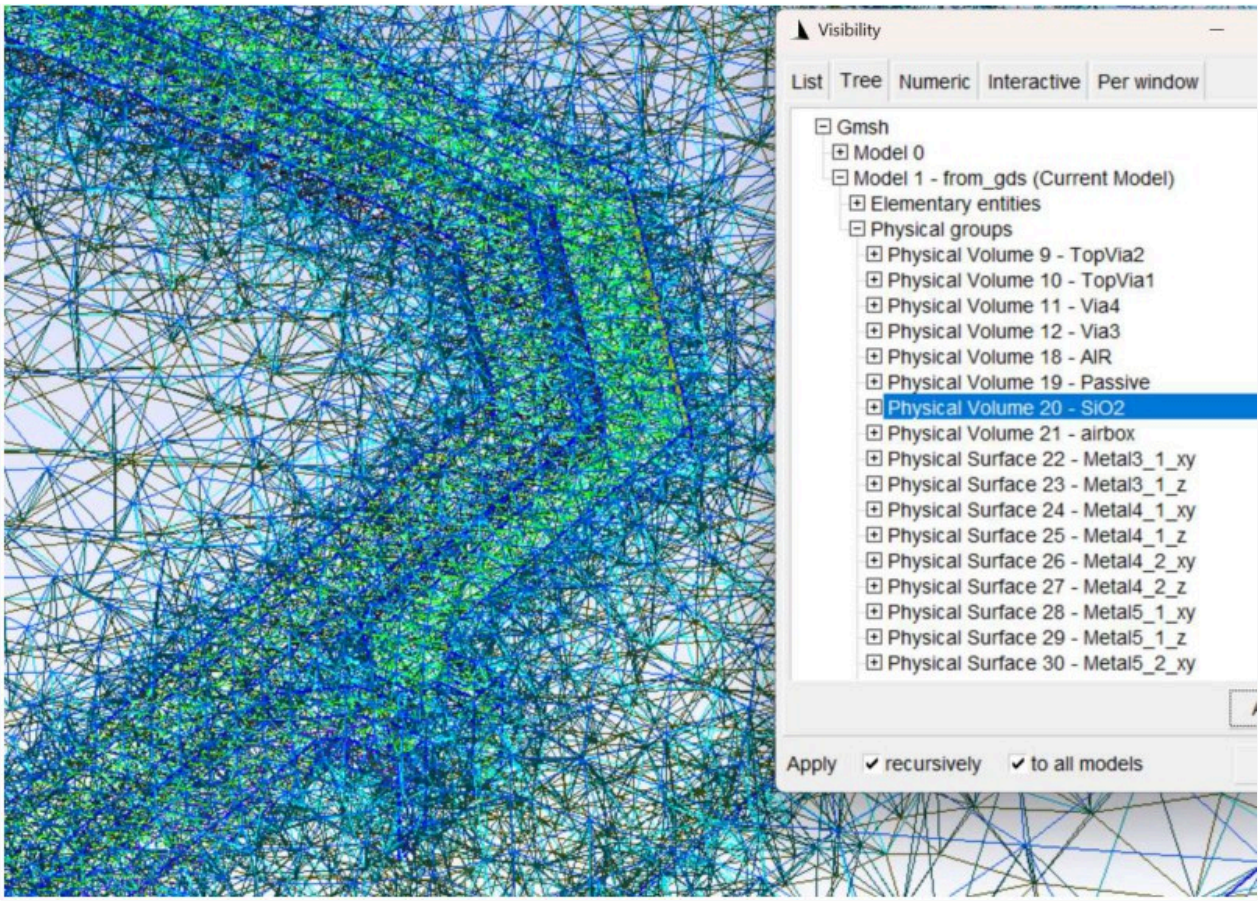


Figure 18: Balun 140-170 GHz: mesh of the SiO2 volume

We will first look at results from refined_cellsize=5 and refined_cellsize=2, both with order=2, over the frequency band 100 to 200 GHz, with all 3 port excitations simulated for full S- parameter data.

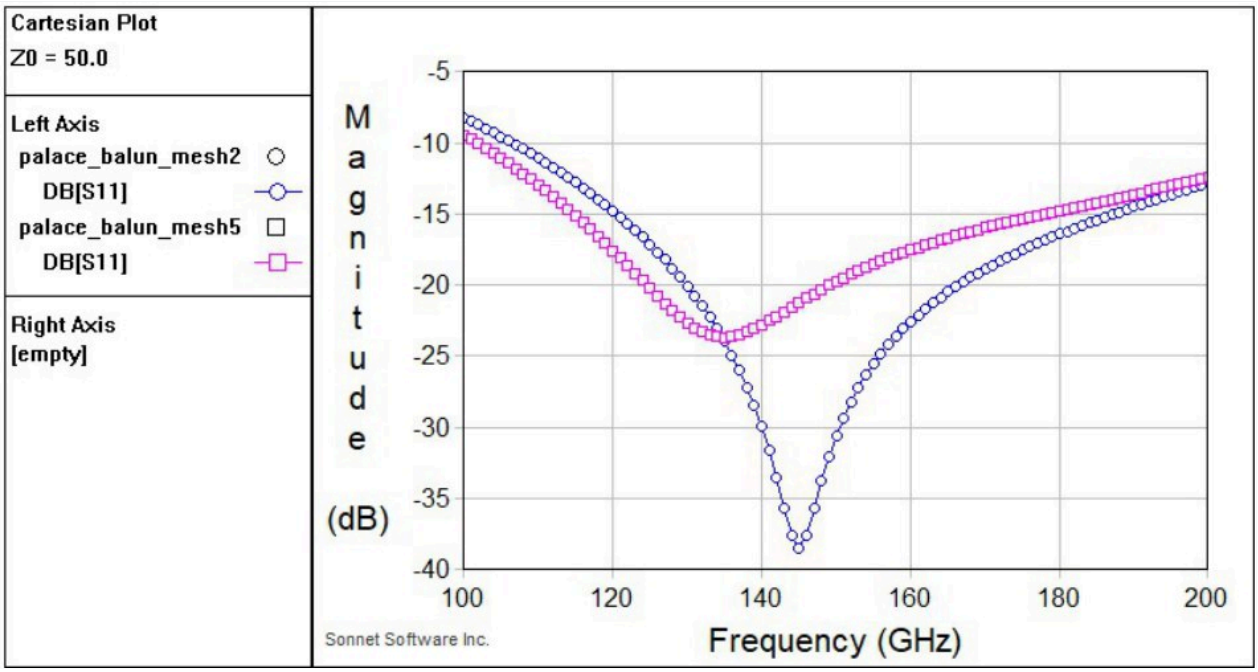


Figure 19: Balun: S11 for refined_cellsize = 5 and 2

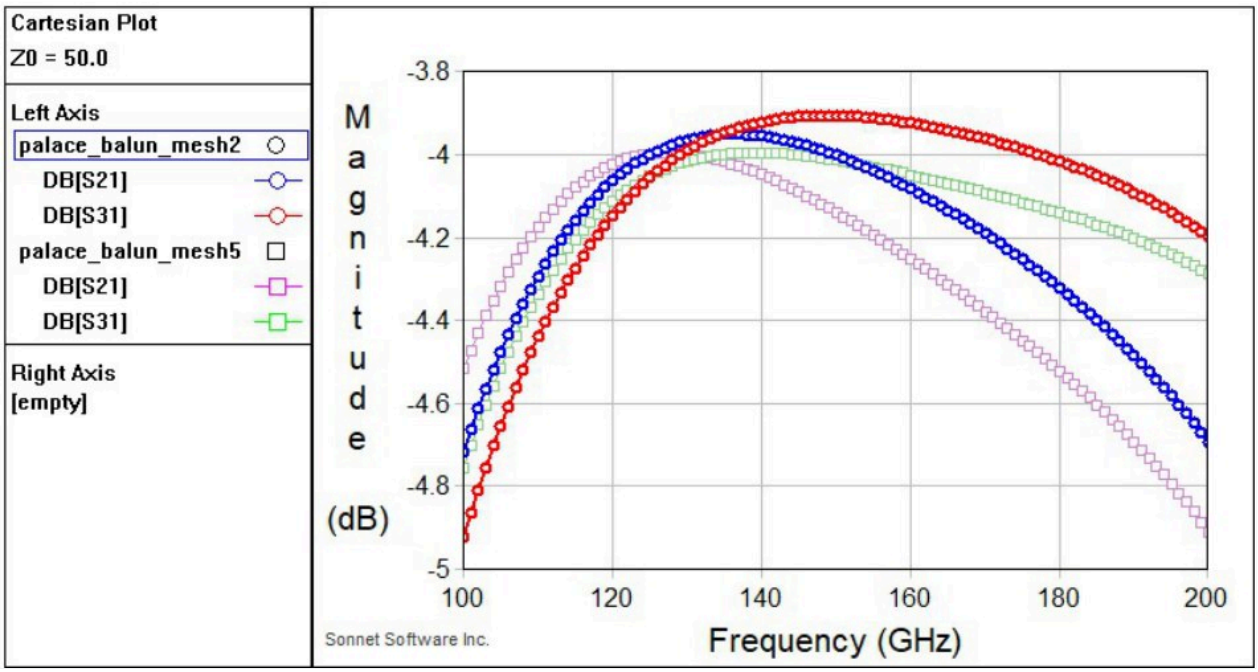


Figure 20: Balun: S21 and S31 for refined_cellsize = 5 and 2

There is quite a difference in S11 and S21,S31 results, and we can assume that the finer mesh gives more accurate results … but how far are we off from convergence? We could now simulate with an ever smaller refined_meshsize = 1, or we can use adaptive mesh refinement now. This will run the simulation starting with an initial mesh size, and then refine the mesh in the most relevant regions.

By default, adaptive mesh refinement in Palace uses data from all simulation frequencies and all port excitations, which means a significant increase in overall simulation time because a full simulation is used during mesh refinement. The excerpt below shows settings from the Palace configuration file for an adaptive sweep with maximum of 2 adaptive mesh refinements, with up to 2 million degrees of freedom.

```
"Model": {
    "Mesh": "palace_balun_mesh5_AMR.msh",
    "L0": 1e-06,
    "Refinement": {
        "UniformLevels": 0,
        "Tol": 0.01,
        "MaxIts": 2,
        "MaxSize": 2000000.0,
        "Nonconformal": true,
        "UpdateFraction": 0.7,
        "SaveAdaptMesh": false
    }
},
"Solver": {
    "Driven": {
        "MinFreq": 100.0,
        "MaxFreq": 200.0,
        "FreqStep": 1.0,
        "SaveStep": 0,
        "AdaptiveTol": 0.01
    },
```

Note for experts: We could decide to create an adaptive mesh refinement at one target frequency, or a few selected target frequencies, and then store that refined mesh for the final full frequency sweep. This is the approach chosen by many commercial FEM solvers, where the user needs to decide what frequencies are used for AMR.

However, that is not implemented in this gds2gmsh workflow. The present implementation of adaptive mesh refinement in gds2gmsh workflow does each refinement step at all frequencies and for all port excitations, to provide a safe and reliable mesh refinement. In this case, the refined mesh exists in memory only and is not stored to disk.

Below is the result from an adaptive mesh refinement with 2 refinement steps, based on an initial mesh with refined_cellsize=5. This result is very close to the results obtained with a (fixed) mesh created from refined_cellsize=2, so both models come to similar results from different (initial) mesh.

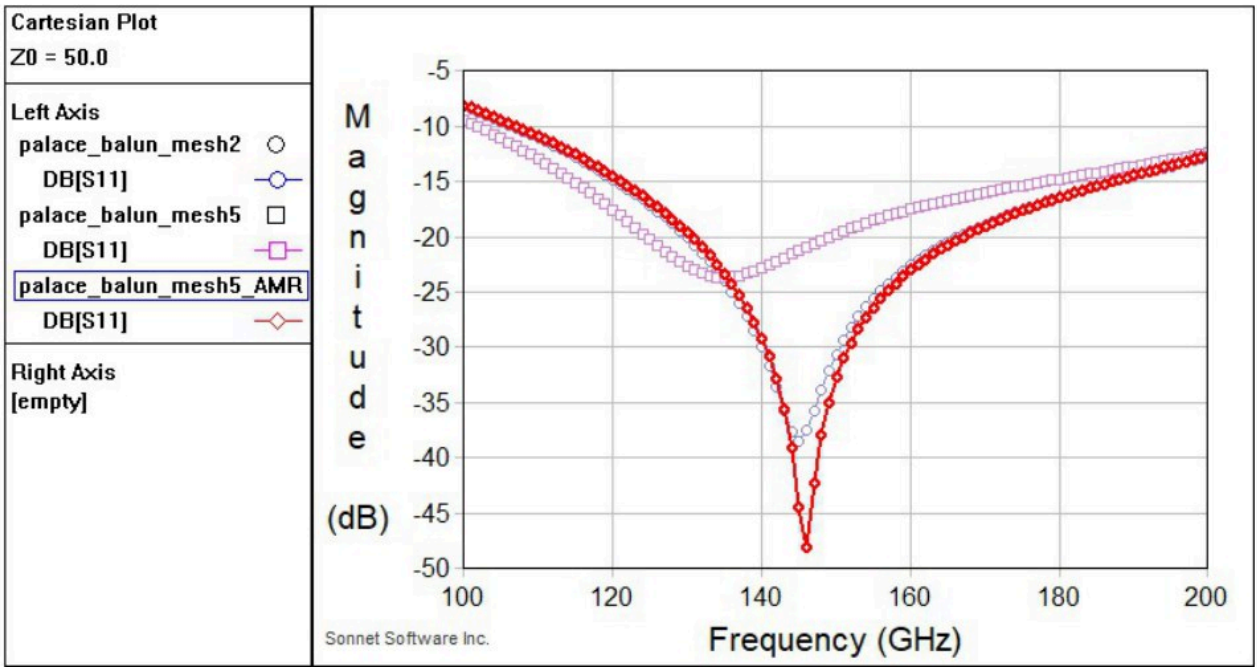


Figure 21: Balun: S11 with adaptive mesh refinement, compared to the fixed meshes

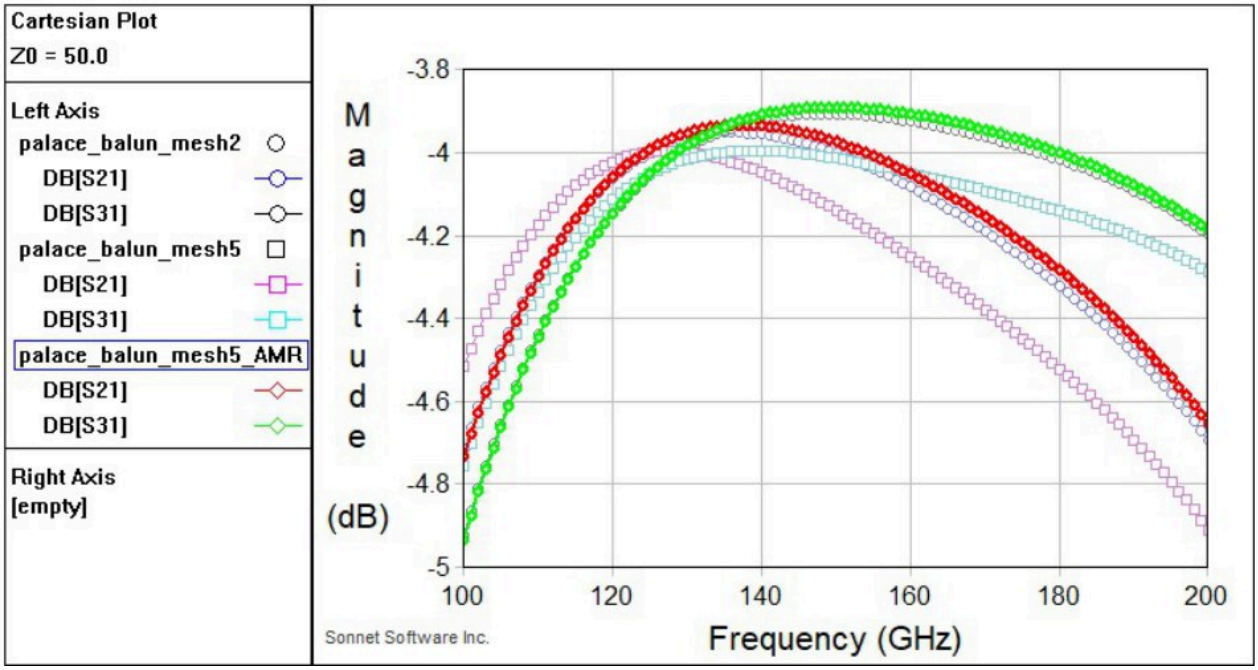


Figure 22: Balun: S21 and S31 with adaptive mesh refinement, compared to the fixed meshes

Here is the overview of simulation times and mesh size for full sweep over all 3 port excitations:

Simulation time total, Degrees of freedom (from log)
refined_cellsize=2: 580s (13 freq), DOF 836878

refined_cellsize=5: 218s (13 freq), DOF 353056
refined_cellsize=5 + AMR 2x: 1132s (13 freq), DOF 353056, 402138,710236

We could now spend more time to double check result with an adaptive mesh refinement starting from an initial mesh that is even finer, and to save time, that could be done at one or few frequencies only. Below is the result for 3 selected frequencies with 2 mesh refinements starting from 2µm:

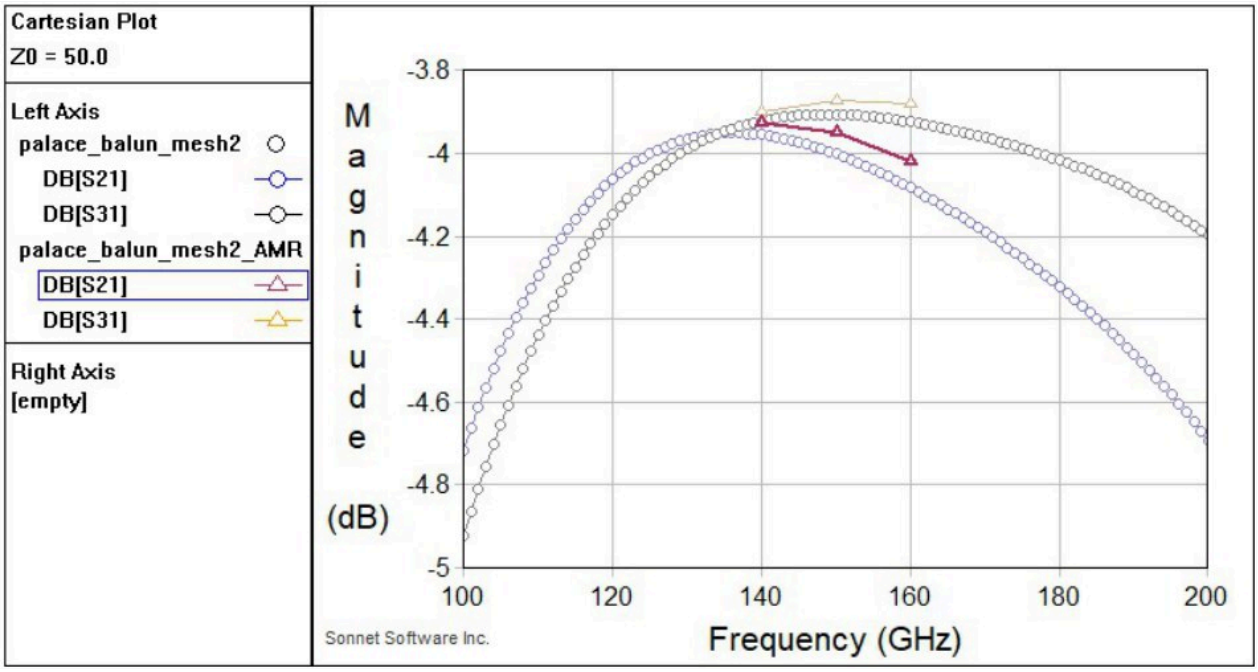


Figure 23: Balun: S21 and S31 with adaptive mesh refinement from the finer initial mesh, at selected frequencies

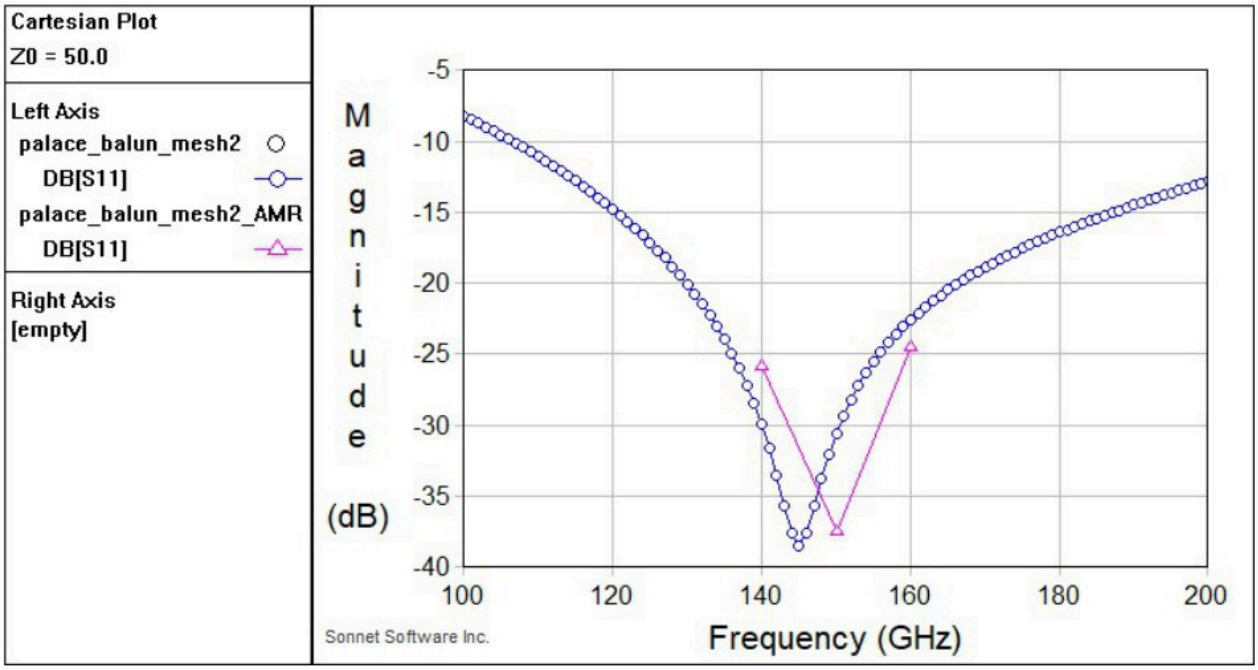


Figure 24: Balun: S11 with adaptive mesh refinement from the finer initial mesh, at selected frequencies

This gives an idea where a converged result would be located in frequency, at approximately 150 GHz resonance instead of 145 GHz calculated with coarser mesh. Here is the overview of simulation times and mesh size for 3 frequencies (not full sweep!) over all 3 port excitations:

Simulation time total, Degrees of freedom (from log)
refined_cellsize=2 + AMR 2x: 1346s (3 freq), DOF 804166, 961866, 1742956

9 Conductor loss modelling

Many RF FEM solvers, including this gds2palace workflow, model metals as hollow bodies with surface resistance on the side walls. The advantage of this approach is that we don't need to mesh into skin effect, which might require sub-micron mesh size inside the conductors.

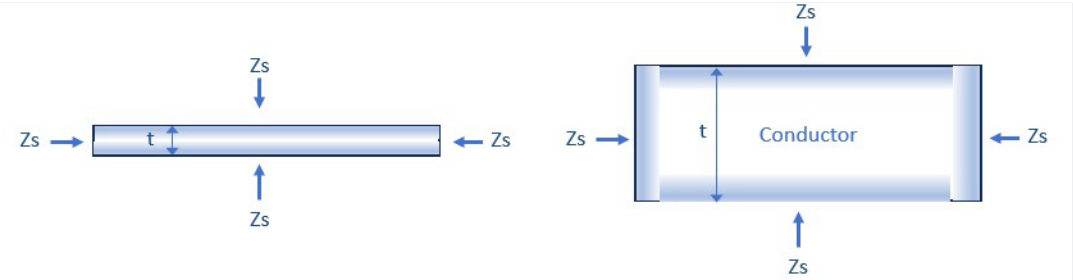


Figure 25: Metals modelled as hollow bodies with surface impedance Zs on all sides

The surface impedance can be calculated easily when physical metal dimensions are much larger than skin depth. However, at low frequency and in the transition between skin effect regime and low frequency, we also need to consider conductor dimensions.

9.1 Limits of conductor loss calculation

The gds2palace workflow creates surface elements for planar metals, and defines them as a conductor sheet with conductivity and thickness in the Palace config.json file. Thickness in this case is the value defined for that metal in the XML stackup file, e.g. 3µm thickness for TopMetal2.

The code in Palace does not know anything about the aspect ratio of the conductor, and uses a surface impedance calculation that is accurate for wide sheets, where we have a conductor that is much wider than the conductor height. Summing up the surface impedance mapped to top and bottom side of a thin sheet, the resulting impedance would be accurate down to DC where it is limited by physical conductor cross section.

However, in our RFIC use case, we might have conductor aspect ratio that reaches almost square shape, and we need to account for side wall currents, not only for closely spaced coupled lines.

Mapping the surface impedance (as calculated internally by Palace from conductor thickness) onto all surfaces, i.e. side walls as well as top and bottom, will lead to an overestimate of available conductor cross section at low frequency:

The Palace calculation[2] of Zs is exact when applied to top and bottom side only. If we apply this value to all sides, the extreme case at DC will over-estimate the conductor cross section by factor (width+thickness)/width.

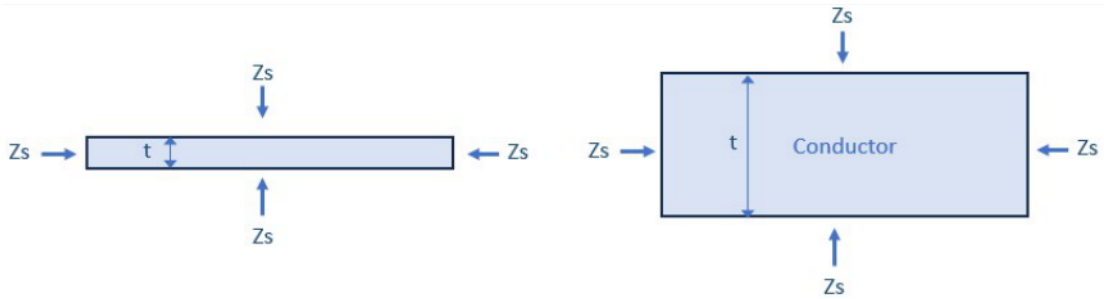


Figure 26: Surface impedance for a thin sheet (left) and a conductor with near-square cross section (right)

The volume models of `settings['conductor_model']` (`'solve_inside'`, `'solve_inside_skin'`) do not have this error, see "Choosing the conductor model" below. An earlier workaround, `settings['z_thickness_factor']`, reduced the thickness used on the side walls. It was removed in

[2] https://github.com/awslabs/palace/blob/main/palace/models/surfaceconductivityoperator.cpp

gds2palace 0.9.0: Palace's thickness correction then changed the side wall impedance well into the skin effect regime, not only at low frequency, and the full thickness was closer to openEMS and measured data (see below).

9.2 Transmission line compared with openEMS and measurement

The 15 µm wide TopMetal2 line over Metal1 ground (`palace_line_viaport.py`) can be compared with an openEMS simulation and with measurement. The Palace results in this section use the default surface impedance model, which overestimates the loss of the Metal1 ground plane, see "Accuracy at high frequency" below.

Palace simulations for this line were done without the substrate below, the stackup didn't include the silicon substrate because we expect the Metal1 ground plane to shield the substrate from the fields in the transmission line. It was verified that the difference with/without substrate below is negligible for these line models.

For openEMS (red curve), the simulation was also done without substrate below, using metal (PEC) side walls. Mesh refinement was set to 0.5µm because openEMS needs to mesh into skin effect. The measurement (light blue) was obtained by de-embedding the line from the overall measurement data with GSG pads, so the overall ripple might be partially due to this more complex setup.

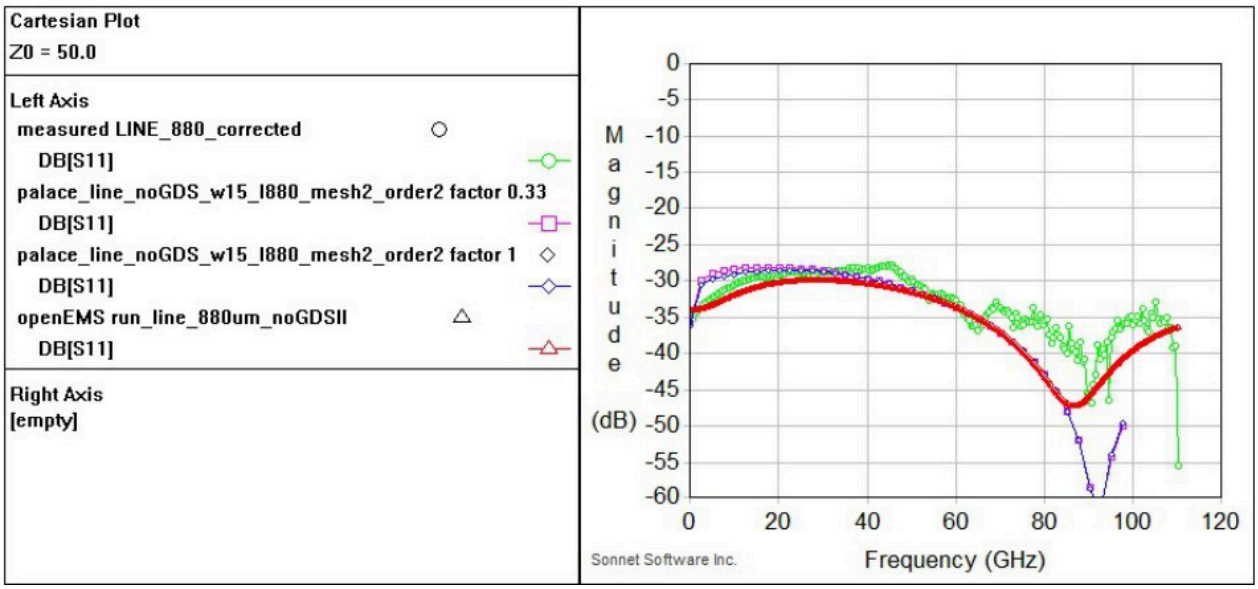


Figure 27: 880 µm line: S11 from Palace, openEMS and measurement

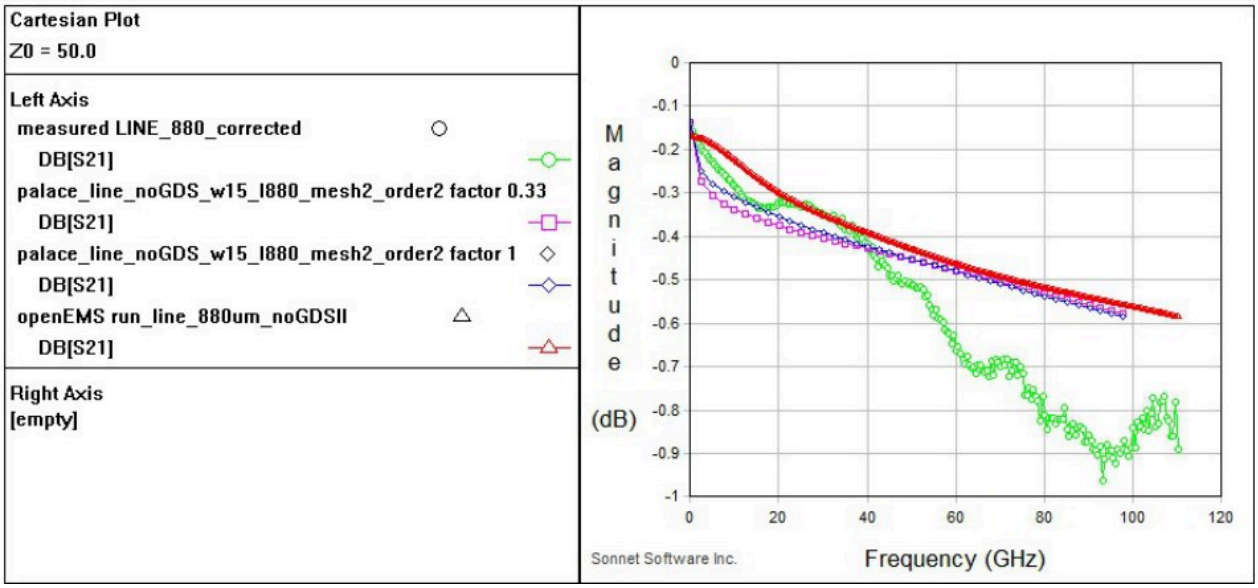


Figure 28: 880 µm line: S21 from Palace, openEMS and measurement

In S21, Palace with the full metal thickness on the side walls (blue) is closer to openEMS and measurement than with the side wall thickness reduced to 1 µm (pink, the removed `z_thickness_factor` = 0.33).
We can also compare data for a longer line of 2580µm, width 15µm TopMetal2 over Metal1.

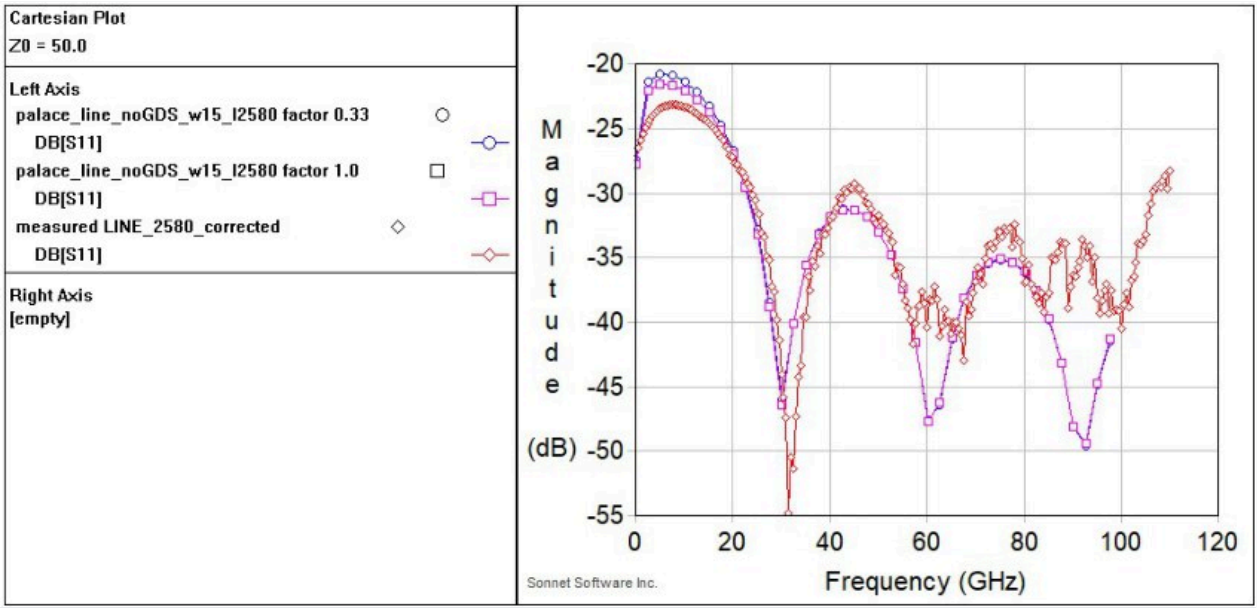


Figure 29: 2580 µm line: S11 from Palace and measurement

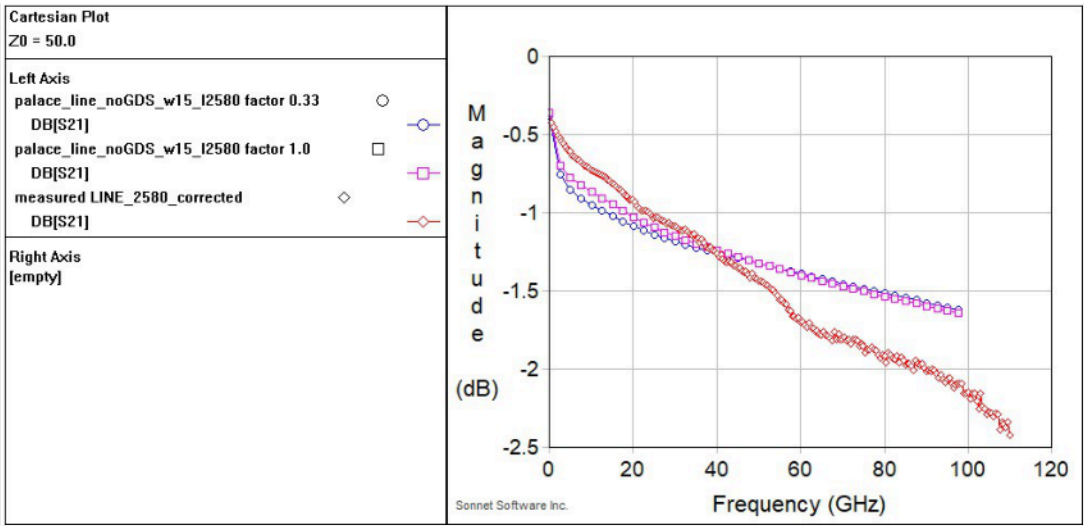


Figure 30: 2580 µm line: S21 from Palace and measurement

Palace with the full side wall thickness is closer to measured data up to 40 GHz; above, the side wall thickness has very little effect.
It can also be seen that measured insertion increases more with frequency than simulated. One possible explanation from an earlier paper was dielectric loss in the SiO2. This dielectric was modelled here as lossless dielectric, loss tangent = 0.
If you prefer to use another value for loss tangent, that is fully supported by the gds2palace workflow, and can be specified easily in the XML stackup file.

9.3 Accuracy at high frequency

Both conductor models have errors at high frequency, of different kinds:

- **Surface impedance** (default): the conductor loss is too low, also well above the frequency range of the side wall effect described above. Each face gets an impedance that is exact for a wide sheet, but near the edges and corners of a conductor the current crowds into a region where the skin layers of two faces overlap, and

a surface impedance per face cannot describe that. For TopMetal2 lines of 2-20 µm width, the resistance is still 7-22 % too low at 50-150 GHz.

- **Surface impedance on thin conductors with current on one face** (default): the conductor loss is too high. Both faces of a conductor share its thickness (see "Palace's External option" below), which assumes that both carry the same current. Where the field reaches only one face, as in a Metal1-Metal5 ground plane under a line, that face carries all the current with twice the sheet resistance of the layer: up to twice the loss of that layer, for layers thinner than about 2 skin depths (in SG13G2: Metal1-Metal5 up to 150 GHz, TopMetal1 below about 9 GHz, TopMetal2 below about 4 GHz). In `palace_line_viaport.py` (TopMetal2 line over a Metal1 ground plane) the line resistance is 21-41 % too high from 1 to 100 GHz.
- **Filled metals** with `refined_cellsize` 2: the conductor loss is too high: with two cell layers of 1.5 µm through the 3 µm thick TopMetal2, the order 2 elements follow the current decay near the surface quite well, but leave some current too deep in the metal once the skin depth (0.3-0.5 µm above 20 GHz) is a fraction of the cell size; the resistance is then 10-20 % too high. What matters is the cell size normal to the conductor surface. A small `refined_cellsize` reduces the error (0.5 µm, i.e. cells of up to 2 skin depths at 100 GHz: +3 % for the inductor below and up to +7 % for `palace_line_viaport.py` up to 100 GHz), but also makes the mesh fine along the surface, which is expensive. Thin layers at the surface (`skin_layers`, about 0.4 and 1 skin depth at the highest frequency) do better at lower cost; thicker layers of 1 and 2.5 skin depths were tested and gave up to +5.5 % (inductor) and +12 % (line) at 100 GHz. `skin_layers` adds the thin layers only where they are needed.

Single-turn inductor, w = 4.5 µm, TopMetal2, peak Q at 77 GHz, compared with filled metals at `refined_cellsize` 0.125 µm (converged: 0.25 µm changes R by less than 1.4 %):

conductor model	refined_cellsize	R at 30 / 75 / 100 GHz	peak Q	run time (8 cores)
filled_metals (reference)	0.125	1 / 1 / 1	22.2	89 min
filled_metals	0.25	+0.4 / +0.8 / +0.9 %	22.0	34 min
filled_metals	2	+5.7 / +8.1 / +9.4 %	20.4	8 min
filled_metals + skin_layers `[0.1, 0.25]`	2	−0.6 / −0.7 / +1.3 %	22.4	10 min
surface impedance (default)	2	−24 / −18 / −14 %	27.1	6 min

`skin_layers = True` (depths from the skin depth at the highest frequency, here 0.094 and 0.236 µm at 150 GHz) gives the same result. On straight TopMetal2 lines of 4.5 and 20 µm width, compared with an independent 2D calculation, the skin layer models are within −2 to +6 % from 1 to 150 GHz, as close as filled metals at `refined_cellsize` 0.25.

`palace_line_viaport.py`, line resistance from two line lengths compared with a 2D calculation of the TopMetal2 line and the lossy Metal1 ground plane: filled metals with `skin_layers = True` (skin layers in TopMetal2 only, Metal1 is too thin) are within −0.5 to +6.4 % from 1 to 100 GHz, as close as filled metals at `refined_cellsize` 0.5 (up to +6.6 %) at half of its run time (34 vs. 68 min on the same machine); filled metals at `refined_cellsize` 2 are up to 11 % too high. Skin layers in the Metal1 plane as well improve the result by about 1 % and make the simulation 2.6 times slower. The surface model is 21-41 % too high here, see above.

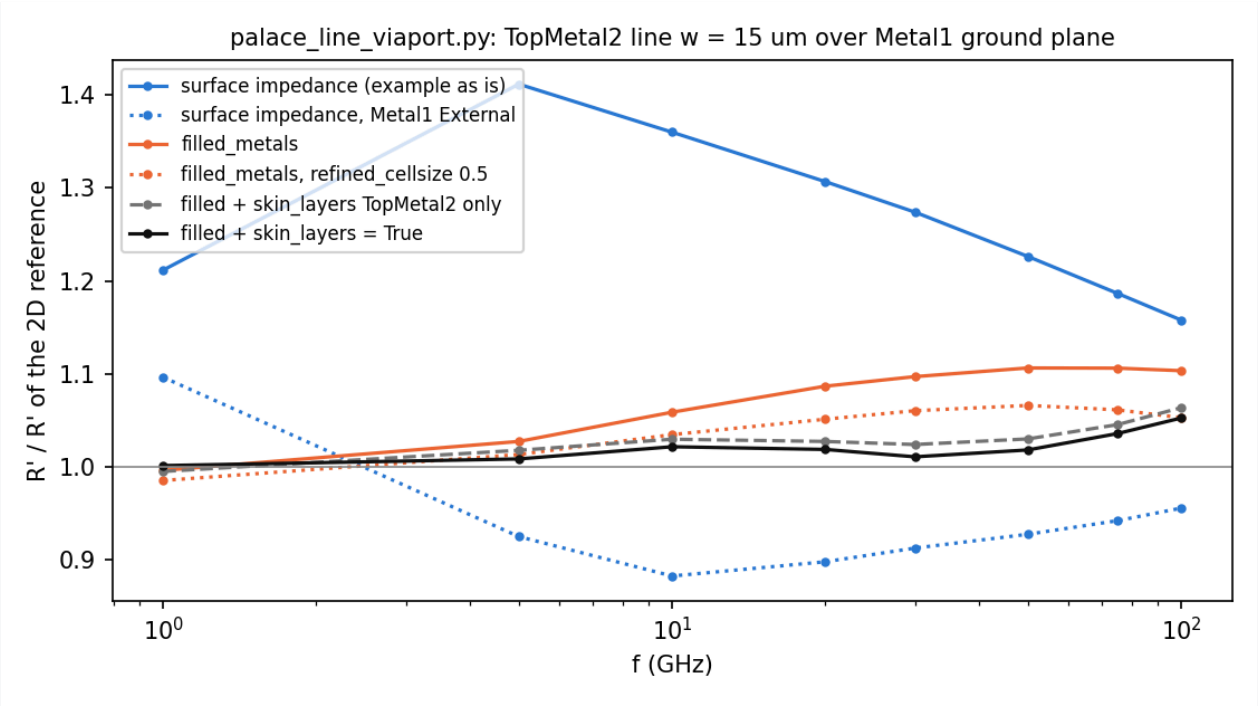


Figure 31: palace_line_viaport.py: line resistance of the conductor models, relative to a 2D calculation

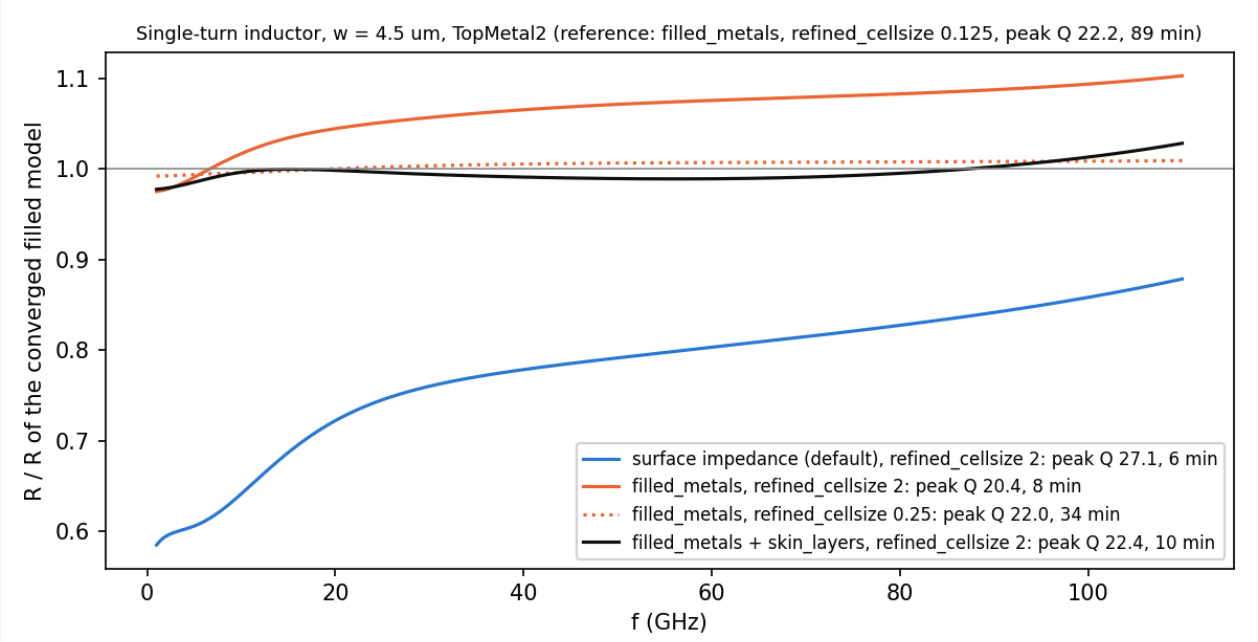


Figure 32: Single-turn inductor: differential resistance of the conductor models, relative to the converged filled model

9.3.1 Choosing the conductor model

`settings['conductor_model']` selects one of three models, named as in HFSS:

	`'surface'` (default)	`'solve_inside'`	`'solve_inside_skin'`
Model	surface impedance on all conductor faces	conductors meshed as volumes	volumes with thin layers along the faces for the skin effect
Comparable to	HFSS impedance boundary (Solve Inside off), Sonnet/Momentum thin metal	HFSS Solve Inside, Sonnet thick metal, Momentum thick conductor	HFSS Solve Inside with a skin depth based mesh operation
Expert settings	`filled_metals = False`	`filled_metals = True`	`filled_metals = True`, `skin_layers = True`
Q of coils on TopMetal1/2	18-30 % too high	4-8 % too low	2-3 % too low
Insertion loss (baluns)	0.02-0.14 dB too low	0.03-0.15 dB too high	up to 0.05 dB too high
Unknowns at the same mesh	1x	about 1.2x	1.75-2.5x
Run time at the same mesh	1x	0.7-2.7x, median 1.4x	1.5-4.7x, median 2x

Errors at a 2 µm mesh, relative to `'solve_inside_skin'` at 1 µm, from step 4 of the mesh convergence studies in `more_examples/mesh_convergence` (five SG13G2 structures). Run times on machines shared with other runs, so they vary by structure and load.

purpose	**`settings['conductor_model']`**
resonance frequencies, inductance, coupling, match and balance	`'surface'`, the fastest
Q and insertion loss during design	`'solve_inside'` (a few percent pessimistic at a 2 µm mesh)
Q and insertion loss, final numbers	`'solve_inside_skin'`

What the errors mean for typical designs (2 µm mesh; reference `'solve_inside_skin'` at 1 µm, for the line a 2D calculation):

Design	Figure of merit	`surface`	`solve_inside`	`solve_inside_skin`
LC VCO tank, coil on TopMetal1/ TopMetal2	coil Q; phase noise ~ $1/Q^2$ where the coil sets the tank Q	Q +18…+30 %: phase noise 1.4-2.3 dB too optimistic	Q −4…-8 %: phase noise 0.35-0.7 dB pessimistic	Q −2…-3 %
LNA input match, baluns, matching networks	insertion loss, adds 1:1 to the noise figure of an LNA	0.02-0.14 dB too optimistic	0.03-0.15 dB pessimistic	up to 0.05 dB pessimistic
MIM capacitor in a tank or matching network	series resistance, capacitor Q	series R about 2x (`palace_rfcmim.py`, 10 MHz: 1.88 instead of 0.87 Ω), capacitor Q about half	right	right
TopMetal2 line over a Metal1 ground plane (`palace_line_viaport.py`)	loss per length	line resistance +21…+41 %	up to +11 %	−0.5…+6.4 %
any structure	inductance, coupling, SRF, match, amplitude and phase balance	all three models within about 1 % of each other		

- **The surface model's loss error has no fixed sign.** It gives too little loss in thick conductors (coils on TopMetal1/TopMetal2) and too much in thin layers that carry current on one face (Metal1-Metal5 ground planes, MIM capacitor plates). On the MIM-loaded balun of the mesh convergence studies, both errors cancelled to 0.02 dB. So a surface model result is not even a bound for the loss.
- **Loss errors in dB hide the relative error.** 0.05 dB on a balun with 1.2 dB insertion loss is 4 % of its loss, 0.1 dB on 1.1 dB is 9 %. At a PA output, 0.1 dB is 2.3 % of the output power.
- **The reference is converged FEM, not measurement.** Against measurement, the stackup matters as much as the conductor model: in the L6n2 inductor study (`more_examples/measured_vs_simulated/more_accurate_models_L6n2_v3`), filled metals at 2 µm with the planar stackup gave peak Q −8.4 % and SRF −9.0 %, with conformal passivation over TopMetal2 Q −4.6 % and SRF −1.6 %, and at 1 µm Q −3.3 % and SRF −1.0 %.

Two more results from the mesh convergence studies:

- Skin layers are sized for the highest simulation frequency, and that costs little accuracy when the sweep goes far beyond the band of interest: skin layers sized for 200 GHz changed the Q of a 30 GHz transformer by 1 %.
- The cell size must not exceed the smallest gap or trace width, for every conductor model. With traces 2 µm apart, `'solve_inside_skin'` at a 4-5 µm mesh was worse than `'solve_inside'` at 2 µm. Where gaps and traces are wider (5 µm in the spiral inductor), `'solve_inside_skin'` at a 5 µm mesh was within 4 % in Q.

9.3.2 Palace's External option

Palace's `Conductivity` boundary uses the surface impedance $Z_s = (\gamma/\sigma)\cdot\coth(\gamma h/2)$ for a conductor of thickness h, which tends to $2/(\sigma h)$ at DC: twice the sheet resistance. This is meant for a conductor that has two faces in the mesh, which share its thickness: a thin sheet drawn as a surface inside the model is split by Palace into two faces ("Added … duplicate boundary elements for interior boundaries" in the Palace output), each gets this Z_s, and in parallel they give the sheet resistance $1/(\sigma h)$. The option `"External": true` (Palace: "Whether this boundary is on the exterior of the computational domain") doubles h, so that Z_s tends to $1/(\sigma h)$: it is meant for a conductor that has only one face in the mesh, on the outer boundary of the model, e.g. a metal enclosure wall or a ground plane that forms the bottom of the model.

gds2palace models each conductor as a hollow body with all its faces in the mesh, the same situation as Palace's split sheet, and does not set `External`. This is exact at low frequency when the top and bottom faces

carry the same current, and makes no difference for conductors much thicker than the skin depth. Real current distributions lie in between, and a boundary condition per face cannot be exact: the current on one face depends on the field at the other face through the metal. Ratio of the real part of Zs with and without `External`:

thickness / skin depth	≤ 0.5	1	1.5	2	2.5	3	4
Re Zs with External / without	0.50	0.54	0.67	0.87	1.03	1.09	1.05

Setting `External` by hand on the faces of a wide ground plane removes the error above in `palace_line_viaport.py`, but it would halve the low frequency loss of conductors that carry current on both faces, e.g. Metal1 lines, and gds2palace groups the top and bottom faces of a conductor together. For thin layers that carry current mainly on one face, `filled_metals` is the better choice: a layer thinner than 2 skin depths is only one or two cells thick, so filling it costs little, and the current distribution in it is then right. `filled_metals = 'auto'` does this for all layers thinner than `refined_cellsize` and keeps the thicker layers as surface impedance.

10 Dielectric loss modelling

The transmission line examples in the metal loss chapter showed a difference between simulated and measured insertion loss which increases toward higher frequencies.

As already mentioned in that metal loss chapter, this could indicate higher **dielectric loss** than modelled. In the supplied XML stackup files, SiO2 is modelled as a lossless dielectric. To verify if dielectric loss could be the reason for poor agreement in that frequency range, we repeat the simulation of the 2580 µm line with non-zero loss tangent for the SiO2 layer.

Loss tangent value used in this simulation is **0.01** , which is a rough estimate based on a paper published by IHP authors for SG25H technologies back in 2007:
Korndörfer F, F Sischka Optimization of the Substrate Parameters for EM-simulators MOSAK/ESSDERC/ESSCIRC Workshop (Munich) 14 Sep., 2007

- https://www.mos-ak.org/munich_2007/posters/P05_MOS-AK_Korndoerfer.pdf

TABLE II
SUMMARY OF TEST STRUCTURES AND RESULTS

Test structure	Area in µm²	Measured capacitance @1 MHz in fF	Measured capacitance @10 GHz in fF	Start values		Optimized values	
				ε_r	Loss tangent	ε_r	Loss tangent
M1-M2	5307	270	265	4.1	0	4.27	0.009
M2-M3	5307	250	245	4.1	0	4.37	0.006
M3-M4	5307	245	235	4.1	0	4.14	0.008
M4-M5	5307	85	80	4.1	0	4.28	0.014

Figure 33: Measured and optimized permittivity and loss tangent from the referenced paper

Indeed, that simulation with SiO2 loss tangent of 0.01 (pink curve) gives increased high frequency loss and quite nice agreement with the measured data above 50 GHz, much better than the model with lossless SiO2.

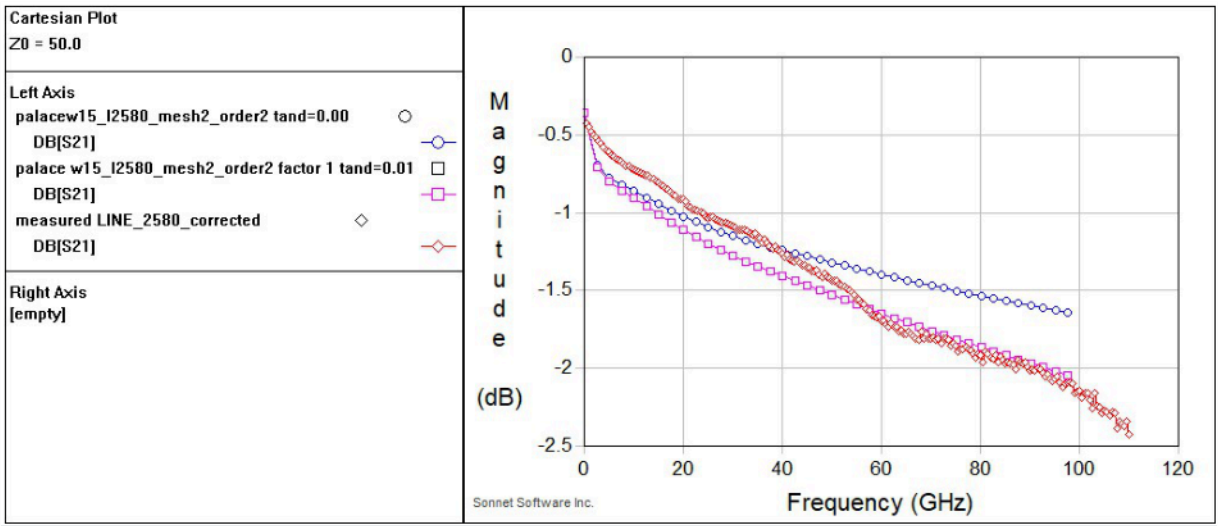


Figure 34: 2580 µm line: S21 with SiO2 loss tangent 0 and 0.01, compared to measurement

For the 880µm line length, the ripple on measurement data has a larger (relative) impact, but we see the same trend: simulation results with tand=0.01 are much closer to measurement than the simulation without SiO2 losses.

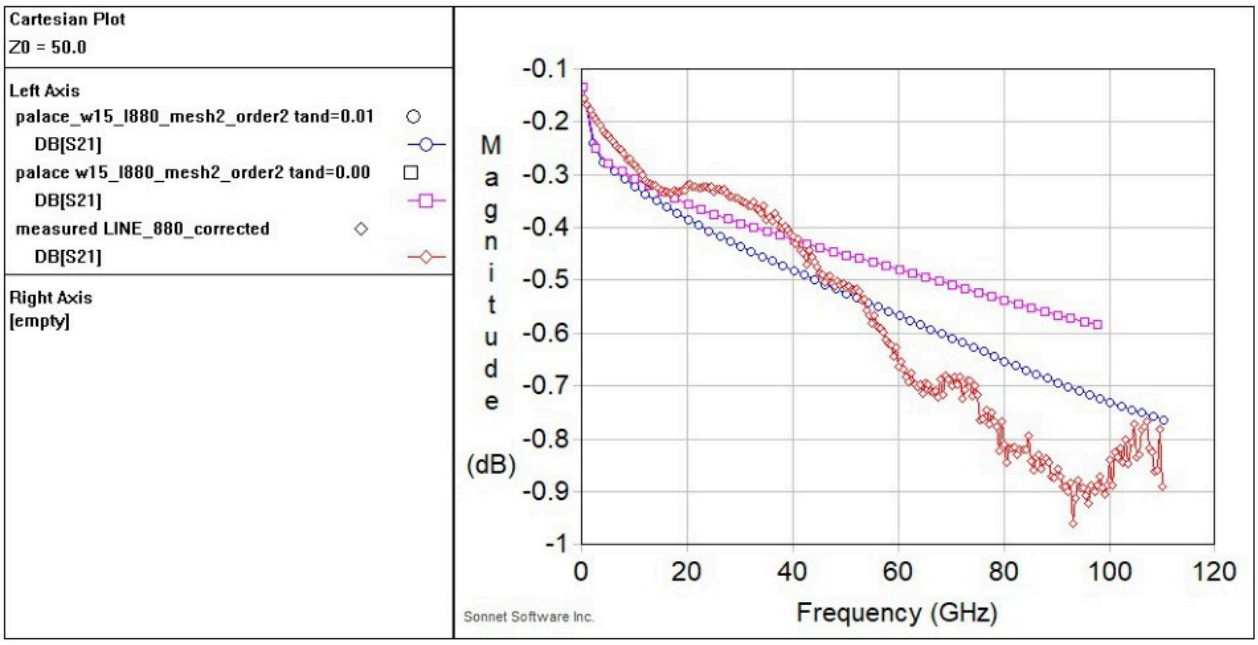


Figure 35: 880 µm line: S21 with SiO2 loss tangent 0 and 0.01, compared to measurement

10.1 Conclusion regarding dielectric losses

The existing data is not complete enough to create an “official” technology stackup with SiO2 losses, but it seems that some amount of dielectric loss is real. If you decide to include dielectric loss in your simulation, this can be easily specified in the XML stackup file using parameter *DielectricLossTangent* for the SiO2 material definition.

11 Advanced topics

11.1 Adaptive mesh refinement at selected frequencies only

By default, adaptive mesh refinement in Palace uses data from all simulation frequencies and all port excitations, which means a significant increase in overall simulation time because a full simulation is used during mesh refinement. To do a two-step approach where the adaptive mesh refinement used only a limited set of frequencies, you could run the gds2palace workflow with those frequencies only, and change the config.json file to **store the resulting mesh data to disk** , by setting SaveAdaptMesh to true.

```
"Model": {
    "Mesh": "palace_balun_mesh2_AMR.msh",
    "L0": 1e-06,
    "Refinement": {
        "UniformLevels": 0,
        "Tol": 0.01,
        "MaxIts": 2,
        "MaxSize": 2000000.0,
        "Nonconformal": true,
        "UpdateFraction": 0.7,
        "SaveAdaptMesh": false
    }
},
```

Adaptive frequency sweep would be **disabled** by removing the “AdaptiveTol” line in the frequency block.

After running Palace with this config file, you will find a refined mesh file in the output folder. Note that the final mesh refinement result is **not** located in the “iteration_nn” folder, the results of adaptive sweep can be found one level above (!) in the parent folder. You would then create another config.json file with the full frequency range and adaptive frequency sweep enabled, with the Mesh setting pointing to that adaptive mesh file created by the AMR before.

11.2 Using wave ports instead of lumped ports not supported

This gds2palace workflow creates lumped ports, which introduce some physical size and thus some parasitic series impedance. The Palace solver also supports using wave ports, but that is not implemented in the gds2gmsh workflow.

12 Using S-Parameter output, model extraction

S-Parameters are an accurate wideband description of the simulation results, but can be difficult to use in time domain circuits simulators. In this chapter, we will look at some possible solutions.

12.1 Lumped circuit model extraction

For a limited range of devices, we can use straightforward calculation of the equivalent lumped circuit model. Such a calculation is available here, based on simple narrowband calculation:

https://github.com/VolkerMuehlhaus/lumpedmodel

Currently, these devices are supported:

- Untapped inductors (2-port)
- MIM capacitor (2-port)
- Transmission line model (RLGC, 2-port data)

These tools are provided as Python scripts. The user needs to specify a target frequency, the fit result is then plotted along with the input data to show the quality of the model fit. For sure, there are other extraction tools available elsewhere that might cover more use cases.

12.2 Mathematical “black box” vector fit

Another approach is to do vector fitting of arbitrary n-port data. One possible implementation is available here, using the vector fit in scikit-rf library: https://github.com/VolkerMuehlhaus/lumpedmodel/tree/main/vector_fit

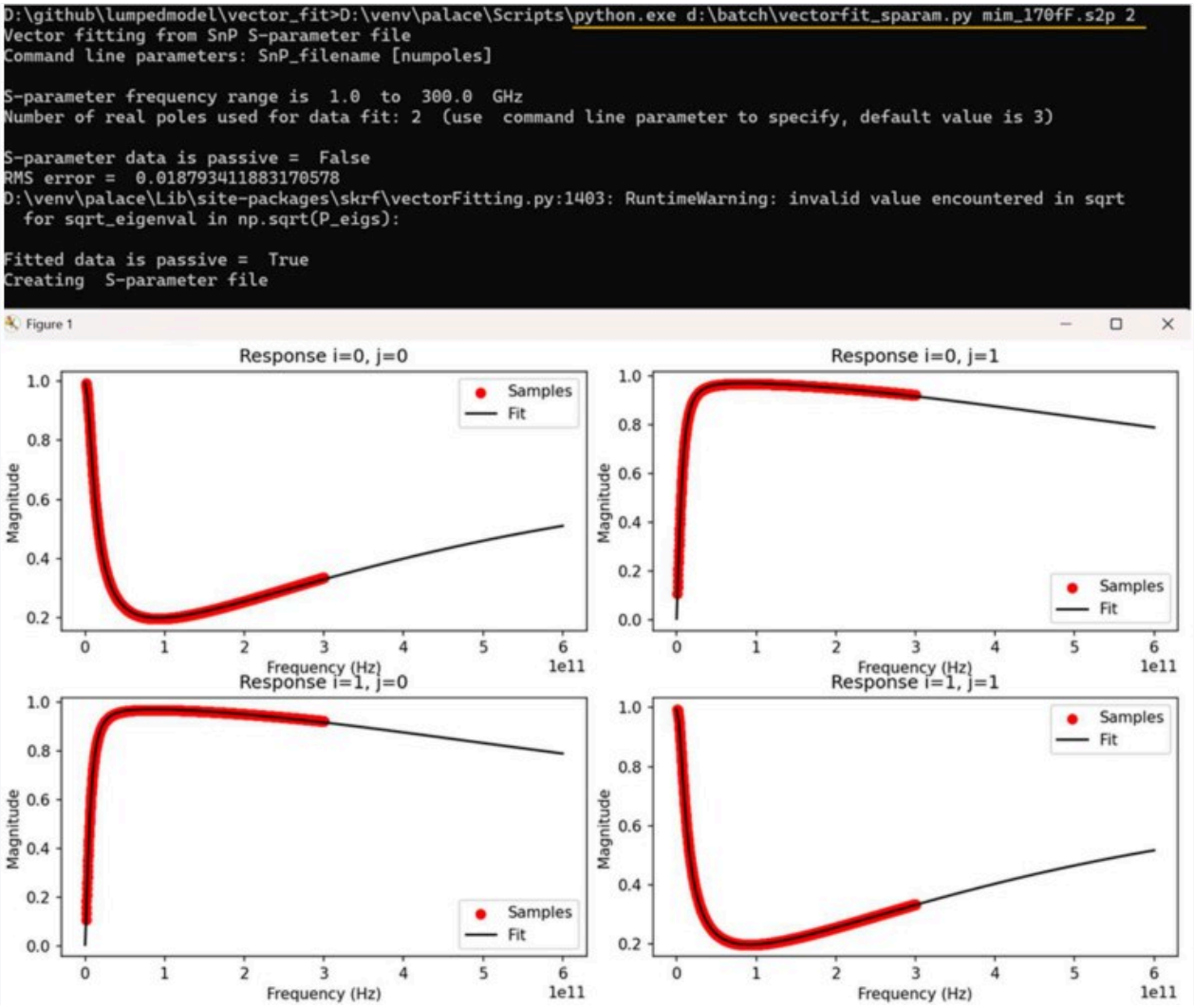


Figure 36: Vector fit of S-parameter data

13 Using gds2palace with Elmer FEM for EM simulation

Elmer FEM is an open-source multiphysics FEM solver. In addition to the AWS Palace output described so far in this guide, gds2palace can create input files for Elmer’s electromagnetic (`VectorHelmholtz`) solver, producing S-parameters from the same layout, stackup and port model already used for Palace.

13.1 Installing Elmer FEM

Elmer FEM is not distributed with gds2palace and must be installed separately, see https://www.elmerfem.org/. gds2palace needs to find two Elmer command-line tools: `ElmerGrid` (mesh conversion) and `ElmerSolver` (the solver itself).

- **Windows:** set the environment variable `ELMER_HOME` to your Elmer install directory. gds2palace looks for `%ELMER_HOME%\bin\ElmerGrid.exe`.
- **Linux/macOS:** make sure `ElmerGrid` and `ElmerSolver` are available on your `PATH`.

If you use `settings['ELMER_MPI_THREADS']` to run Elmer across multiple processes, you also need an MPI implementation installed (see "External tools (not Python modules)" above) — on Windows this is Microsoft MPI, providing the `mpiexec`/`mpirun` launcher the generated `run_elmer` script uses.

13.2 From a Palace model to an Elmer model

An Elmer EM model uses the same **GDSII layout**, **XML stackup file**, **settings dictionary** and **port definitions** already described in chapter "Simulation model file in detail". The only change needed at the end of the model script is which output function is called:

```
# Palace output
config_name, data_dir = simulation_setup.create_palace (excite_ports, settings)
utilities.create_run_script(sim_path)
```

becomes

```
# Elmer FEM output
config_name, data_dir = simulation_setup.create_elmer (excite_ports, settings)
utilities.create_elmer_run_script(sim_path, settings)
```

Everything else in the model script — `gds_filename`, `XML_filename`, `settings['unit']`, `settings['margin']`, `settings['refined_cellsize']`, and port definitions via `simulation_setup.all_simulation_ports()` / `simulation_setup.simulation_port(...)` — stays exactly the same as for a Palace model.

Important difference in how ports are excited: Palace runs one solver pass per active port excitation (controlled by each port's `voltage` and the `excite_ports` list) to build up the full S-matrix, and ports with `voltage=0` are skipped to save simulation time. Elmer's EM solver instead uses a *constraint modes* ("scanning") analysis that solves for **all** ports defined in the model in a single run. This means:

- Every port you define in `simulation_ports` appears in the resulting Elmer S-matrix, regardless of its `voltage` setting.
- There is no equivalent to the Palace "excite only port 1" trick to save time — with Elmer, you get the full n-port S-matrix or nothing.
- Depending on port count and mesh size, a full n-port Elmer run can be faster or slower than n separate Palace excitation runs; this depends on the specific model.

13.3 Frequency sweep

`settings['fstart']`, `settings['fstop']`, `settings['fstep']`, `settings['fpoint']` and `settings['fdump']` are read the same way as for Palace and written to a plain frequency list (`frequencies.dat`) for Elmer. Note that Palace's adaptive frequency sweep interpolation (`settings['adaptive_sweep']`) has no Elmer equivalent — Elmer solves at every frequency you list, so simulation time scales directly with the number of frequency points.

13.4 Mesh order and solver method

```
settings['order'] = 2          # 1 = linear, 2 = quadratic (recommended), default 2
settings['iterative'] = False  # True = iterative linear solver, False = direct solver (default)
```

- `settings['order']` selects the basis function order, same setting as used for Palace. Elmer supports order 1 (linear) and order 2 (quadratic); other values fall back to the order-1 solver recipe.
- `settings['iterative']` selects between Elmer's direct solver (MUMPS, default, robust for small/medium models) and an iterative solver, which can be more memory-efficient for large models.

- `settings['ELMER_MPI_THREADS']` (optional) enables MPI-parallel solving: gds2palace partitions the mesh with `ElmerGrid` for the given number of MPI processes, and the generated `run_elmer` script (see below) uses `mpirun` to start `ElmerSolver`.

```
settings['ELMER_MPI_THREADS'] = 8   # partition mesh and solve using 8 MPI processes
```

13.5 Boundary conditions and current limitations

`settings['boundary']` (the six-sided outer simulation boundary, values ABC/PML/PEC/PMC) works the same way as for Palace, including PMC.

A few other Palace features are not yet available in the Elmer EM flow:

- **Sheet resistor layers** (`Type="sheet"` metal paired with a `Resistor` material, as used for on-chip resistors) are not yet supported for Elmer EM output.
- **Composite ports** (grouping several EM ports into one output port) are not supported, same limitation as for Palace.

13.6 Running the Elmer EM simulation and getting S-parameters

`utilities.create_elmer_run_script(sim_path, settings)` writes a `run_elmer` script into the model output directory, next to the generated Elmer input files (`physics.sif`, `case.sif`, `frequencies.dat`, `mesh/`, `ELMERSOLVER_STARTINFO`):

```
#!/bin/bash
ElmerSolver
combine_snp
```

or, if `settings['ELMER_MPI_THREADS']` is set to more than 1:

```
#!/bin/bash
mpirun -np 8 ElmerSolver case.sif
combine_snp
```

Run it from the model output directory:

```
cd <model>_data
./run_elmer
```

`ElmerSolver` finds `case.sif` automatically via `ELMERSOLVER_STARTINFO`. The same `combine_snp` postprocessing script already used for Palace results also recognizes Elmer's `scalar_results` output and converts it to Touchstone (`.sNp`) format, so downstream use of Elmer S-parameters is identical to Palace results.

14 Using gds2palace with Elmer FEM for thermal simulation

In addition to EM simulation, gds2palace can generate input for Elmer's steady-state Heat Equation solver, producing a 3D temperature field from user-defined heat sources and constant-temperature boundaries. This shares the same GDSII + XML stackup pipeline as the EM flows, but does not use frequencies or ports.

Elmer FEM must be installed as described above (`ElmerGrid` and `ElmerSolver` on `PATH`, or `ELMER_HOME` set on Windows).

A complete, extensively documented worked example (model script, stackup XML, and GDSII layout) is available at `more_examples/thermal_simulation_using_Elmer/Elmer_Thermal_Workflow.md` — this chapter summarizes the key points.

14.1 Stackup XML: thermal material properties

For a thermal run, every <Material> needs a thermal conductivity. Only steady-state conductivity is evaluated so far; electrical parameters (used for S-parameter simulation) are not needed for a thermal-only model, though the same stackup file can be reused for both EM and thermal simulation.

```
<Material Name="TopMetal2" Type="Conductor" Conductivity="30300000.0"
          Density="2700" ThermalConductivity="237" Color="ff8000"/>
```

- `Density` (kg/m³) is written through to Elmer but not used by the steady-state solver.
- `ThermalConductivity` is a single value in W/(m·K), for materials whose conductivity doesn't change meaningfully with temperature.
- `ThermalConductivityTable` points to a <Table> instead, for temperature-dependent conductivity (e.g. silicon substrate):

```
<Material Name="HighResSubstrate" Type="Semiconductor" Conductivity="0.025"
          ThermalConductivityTable="SiliconThermalCond" Density="2329"/>
...
<Tables>
  <Table Name="SiliconThermalCond">
    <Point Temperature="280" Value="163.00"/>
    <Point Temperature="290" Value="155.20"/>
  </Table>
</Tables>
```

gds2palace writes this straight through as an Elmer temperature-dependent material table — no manual `.sif` editing needed. See chapter "Extended XML stackup format" in the Appendix and `XML_stackup_format.md` for the full attribute reference.

14.2 Heat sources and constant-temperature boundaries

Instead of ports, a thermal model declares where heat enters and leaves through `thermal_objects`, using ordinary marker polygons drawn on dedicated GDSII layers:

```
thermal_objects = simulation_setup.all_thermal_objects()
thermal_objects.add_heatsource(simulation_setup.heatsource(
    power=0.65, source_layernum=201, target_layername='TFR'))
thermal_objects.add_consttemp(simulation_setup.constanttemp(
    temp=298, source_layernum=202, target_layername='BACKSIDEGND'))
```

- `source_layernum` is a GDSII layer number you draw purely as an xy footprint marker; it does not set a z-position.
- `target_layername` is the name of an existing <Layer> from the stackup XML — its z-range becomes the 3D volume the source/boundary is applied to.
- `heatsource(power, ...)` dissipates `power` Watts as a volumetric heat source over that volume. In the example above, 0.65 W is applied to the `TFR` resistor layer via marker layer 201.
- `constanttemp(temp, ...)` fixes that volume's temperature (Kelvin), acting as a heat sink/reference boundary. In the example above, layer 202 pins the `BACKSIDEGND` layer to 298 K (25 °C), a typical backside heatsink boundary.

The target layer for a heat source must be a volume (`Zmax` larger than `Zmin`). The target layer for a constant-temperature boundary is usually a zero-thickness sheet layer (`Zmax = Zmin`) — if it has finite thickness instead, two boundaries are created, one at each face.

14.3 Model script settings

```
settings['unit']              = 1e-6   # geometry is in microns
settings['margin']            = 100    # air margin around the layout, in microns
```

```
settings['refined_cellsize']  = 5      # extra-fine mesh near heat sources
settings['meshsize_max']      = 100    # coarsest mesh size, in microns
settings['elmer_thermal']     = True   # selects the thermal flow instead of EM/Palace
settings['thermal_objects']   = thermal_objects
```

`settings['no_gui'] = True` skips the interactive gmsh mesh preview, useful for unattended/batch runs. Then create the model:

```
config_name, data_dir = simulation_setup.create_elmer_thermal (settings)
```

Note there is no `excite_ports` argument here — a thermal run has no ports.

14.4 Running ElmerSolver and viewing results

Output goes to an `elmer_model/` folder next to the script, containing the gmsh mesh, the Elmer-native `mesh/` folder (from `ElmerGrid`), `case.sif` (materials, body/volume definitions, the heat source as a volumetric Body Force, and the constant-temperature Boundary Condition), and `ELMERSOLVER_STARTINFO`. It's worth a quick look at `case.sif` before solving, especially the first time you use a new stackup — it's plain text and easy to sanity-check.

```
cd elmer_model
ElmerSolver
```

Results appear in the same directory:

- `thermal_results.vtu` — full 3D temperature field, open in ParaView.
- `thermal_results.dat` — quick min/max temperature summary (plain text).

If `settings['ELMER_MPI_THREADS']` is set to more than 1, gds2palace also partitions the mesh for you; run `ElmerSolver_mpi` with your MPI launcher of choice instead of the single-threaded `ElmerSolver`.

14.5 Using setupThermal instead of writing Python code

setupThermal is a desktop GUI (part of the `setupEM` project) that drives this same flow without writing a Python model script by hand: pick the layout/stackup files, add heat sources and constant-temperature boundaries in a table (GDSII source layer, target stackup layer name, and power in W or temperature in K), then generate and run the model, or launch `ElmerSolver` directly from the app with output streamed to an on-screen log. The **Model** tab shows the generated Python code, so you can inspect or export it if you'd rather take over from there manually.

15 Appendix

15.1 Understanding volumes and surfaces created from GDSII

The workflow creates different types of geometry: volumes and surfaces. This depends on the mapping in the XML stackup file.

Geometries from GDSII layers are defined in the "Layers" section of the XML file :

- Type="conductor" is used for all regular metal layers, in the mesh they are represented by **"Surface"** objects on all boundaries (top, bottom, side walls). In other words, these metals are represented by hollow shells and not by solid volumes.
- Type="sheet" is used for metal layers that have no physical height in the model, they are represented as one flat **"Surface** " only.
- Type="via" is used for via layers, these are represented in the mesh by **"Volume"** objects.

Important: Conductor layers must never be stacked directly, they must be separated by a via layer!

Extract from an XML stackup file, showing layer mappings with conductor, via and sheet types:

```
  </Dielectrics>
  <Layers>
    <Substrate Offset="183.75"/>
    <Layer Name="Activ" Type="conductor" Zmin="0.0000" Zmax="0.4000" Material="Activ" Layer="1"/>
    <Layer Name="Metal1" Type="conductor" Zmin="1.0400" Zmax="1.4600" Material="Metal1" Layer="8"/>
    <Layer Name="Metal2" Type="conductor" Zmin="2.0000" Zmax="2.4900" Material="Metal2" Layer="10"/>
    <Layer Name="Metal3" Type="conductor" Zmin="3.0300" Zmax="3.5200" Material="Metal3" Layer="30"/>
    <Layer Name="Metal4" Type="conductor" Zmin="4.0600" Zmax="4.5500" Material="Metal4" Layer="50"/>
    <Layer Name="Metal5" Type="conductor" Zmin="5.0900" Zmax="5.5800" Material="Metal5" Layer="67"/>
    <Layer Name="TopMetal1" Type="conductor" Zmin="6.4303" Zmax="8.4303" Material="TopMetal1" Layer="126"/>
    <Layer Name="TopMetal2" Type="conductor" Zmin="11.2303" Zmax="14.2303" Material="TopMetal2" Layer="134"/>
    <Layer Name="TopVia2" Type="via" Zmin="8.4303" Zmax="11.2303" Material="TopVia2" Layer="133"/>
    <Layer Name="TopVia1" Type="via" Zmin="5.5800" Zmax="6.4303" Material="TopVia1" Layer="125"/>
    <Layer Name="Via4" Type="via" Zmin="4.5500" Zmax="5.0900" Material="Via4" Layer="66"/>
    <Layer Name="Via3" Type="via" Zmin="3.5200" Zmax="4.0600" Material="Via3" Layer="49"/>
    <Layer Name="Via2" Type="via" Zmin="2.4900" Zmax="3.0300" Material="Via2" Layer="29"/>
    <Layer Name="Via1" Type="via" Zmin="1.4600" Zmax="2.0000" Material="Via1" Layer="19"/>
    <Layer Name="Cont" Type="via" Zmin="0.4000" Zmax="1.0400" Material="Cont" Layer="6"/>
    <Layer Name="SUBGND" Type="conductor" Zmin="-3.75" Zmax="0" Material="LOWLOSS" Layer="210"/>
    <Layer Name="BACKSIDEGND" Type="sheet" Zmin="-180" Zmax="-180" Material="LOWLOSS" Layer="211"/>
    <Layer Name="MIM" Type="conductor" Zmin="5.6043" Zmax="5.7540" Material="MIM" Layer="36"/>
    <Layer Name="Vmim" Type="via" Zmin="5.7540" Zmax="6.4303" Material="Vmim" Layer="129"/>
    <Layer Name="LBE" Type="dielectric" Zmin="-183.75" Zmax="0" Material="Air" Layer="157"/>
  </Layers>
</ELayers>
</Stackup>
```

Resulting mesh with surface and volume objects:

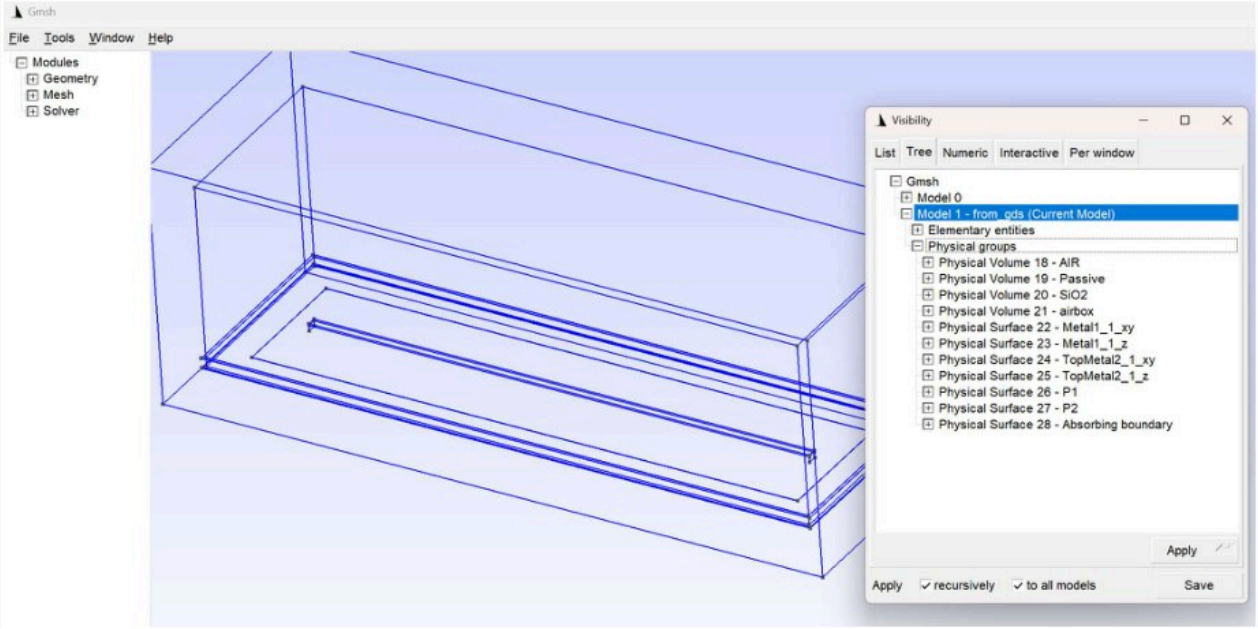


Figure 37: Resulting mesh with surface and volume objects

Dielectrics: There are additional materials in the stackup that are not drawn in the GDSII file, such as substrate or oxide or passivation. These are created as **“Volume”** objects in the mesh.
Dielectrics are defined in the section of the XML file, and usually cover the entire drawing area (bounding box of GDSII drawing) plus an additional “margin” that is defined in the simulation model code.

```
</Materials>
<ELayers LengthUnit="um">
  <Dielectrics>
    <Dielectric Name="AIR" Material="AIR" Thickness="200.0000"/>
    <Dielectric Name="Passive" Material="Passive" Thickness="0.4000"/>
    <Dielectric Name="SiO2" Material="SiO2" Thickness="15.7303"/>
    <Dielectric Name="EPI" Material="EPI" Thickness="3.7500"/>
```

```
    <Dielectric Name="Substrate" Material="Substrate" Thickness="180.0000"/>
  </Dielectrics>
  <Layers>
    <Substrate Offset="183.75"/>
    <Layer Name="Activ" Type="conductor" Zmin="0.0000" Zmax="0.4000" Material="Activ" Layer="1"/>
    ...
```

As an option, the size of these dielectric layers can be defined by an optional parameter **Boundary** which sets the size of the dielectric from the bounding box of that layer in the GDSII file.

```
</Materials>
<ELayers LengthUnit="um">
  <Dielectrics>
    <Dielectric Name="iAIR" Material="AIR" Thickness="50.0000" Boundary="1039"/>
    <Dielectric Name="iSubstrate" Material="Substrate" Thickness="80.0000" Boundary="1039"/>
    <Dielectric Name="iSiO2" Material="SiO2" Thickness="12.27" Boundary="1039"/>
    <Dielectric Name="iPassive" Material="Passive" Thickness="0.4000" Boundary="1039"/>
    <Dielectric Name="Underfill" Material="Underfill" Thickness="46.2" Boundary="1039"/>
    <Dielectric Name="Passive" Material="Passive" Thickness="0.4000" Boundary="39"/>
    <Dielectric Name="SiO2" Material="SiO2" Thickness="15.7303" Boundary="39"/>
    <Dielectric Name="EPI" Material="EPI" Thickness="3.7500" Boundary="39"/>
    <Dielectric Name="Substrate" Material="Substrate" Thickness="180.0000" Boundary="39"/>
    <Dielectric Name="AIR" Material="AIR" Thickness="50.0000" Boundary="39"/>
  </Dielectrics>
  <Layers>
    <Substrate Offset="233.75"/>
    <Layer Name="Activ" Type="conductor" Zmin="0.0000" Zmax="0.4000" Material="Activ" Layer="1"/>
    ...
```

15.2 Mapping of Volumes and Surfaces to Palace materials

Palace offers a wide range of options to assign material properties to volumes and surfaces. Our workflow creates that mapping in the Palace control file (config.json), as described below:

Via layers (Type="via") are created as volumes, and the conductivity defined in the XML stackup is assigned to these volumes. In the Palace configuration file excerpts below, "Attributes" is the number of the physical volume or surface in the mesh.

```
"Domains": {
    "Materials": [
        {
            "Attributes": [
                9
            ],
            "Permittivity": 1.0,
            "Conductivity": [
                314300.0,
                314300.0,
                3143000.0
            ]
        },
```

To account for the z-directed nature of via arrays, which allows current flow predominantly in z direction, the conductivity from XML is only assigned to z-direction, and the value in xydirection is reduced by a factor of 10. This avoids issues with "unreal" currents flowing on the side walls of merged via polygons after via array merging.

With settings['fill_factor_correction'] = True, the conductivity of merged via polygons is additionally multiplied by their fill factor, in all directions. Via polygons on the same layer with different fill factors then get separate materials, in physical groups named after the layer and the fill factor, e.g. TopVia2_x0.49. The same

applies to Elmer output: Elmer EM scales the (scalar) electrical conductivity, Elmer thermal the heat conductivity, and the Elmer bodies are named after these physical groups.

Metal layers (Type="conductor") are created as hollow elements surrounded by surfaces, with surface impedance to define metal loss (default `conductor_model = 'surface'`). The conductivity and thickness are obtained from the XML file. With `'solve_inside'` or `'solve_inside_skin'`, they are volumes with the conductivity of the XML file instead, like the vias.

```
"Boundaries": {
    "Conductivity": [
        {
            "Attributes": [
                27
            ],
            "Conductivity": 21640000.0,
            "Thickness": 0.41999999999999993
        },
```

All faces, top, bottom and side walls, get the same conductivity and thickness. This is exact for top and bottom of wide conductors, down to low frequency where the skin depth is larger than the physical conductor height. On the side walls, it over-estimates the conductor cross section at low frequency, see "Limits of conductor loss calculation".

Metall sheet layers (Type="sheet") are created as a single, flat surface. Their position in the stackup should be defined with Zmin=Zmax for better readability, but actually the Zmax value is ignored here.

```
<Layer Name="MIM" Type="conductor" Zmin="5.6043" Zmax="5.7540" Material="MIM" Layer="36"/>
<Layer Name="Vmim" Type="via" Zmin="5.7540" Zmax="6.4303" Material="Vmim" Layer="129"/>
<Layer Name="LBE" Type="dielectric" Zmin="-183.75" Zmax="0" Material="Brick" Layer="157"/>
<Layer Name="ResistorA" Type="sheet" Zmin="22.0" Zmax="22.0" Material="ResA" Layer="220"/>
<Layer Name="ResistorB" Type="sheet" Zmin="15.0" Zmax="15.0" Material="ResB" Layer="221"/>
```

These sheets are typically used with a Type="Resistor" material definition in the XML file, with fixed sheet resistance in Ohm/square.

```
<Material Name="Vmim" Type="Conductor" Permittivity="1" DielectricLossTangent="0" Conductivity="2191000.0"
Color="ffe6bf"/>
<Material Name="MIM" Type="Conductor" Permittivity="1" DielectricLossTangent="0" Conductivity="500000.0"
Color="e6ffbf"/>
<Material Name="LBE" Type="Dielectric" Permittivity="1.0" DielectricLossTangent="0.0" Conductivity="0"
Color="d0d0d0"/>
<Material Name="ResA" Type="Resistor" Rs="3.5e-1" Color="d0d0d0"/>
<Material Name="ResB" Type="Resistor" Rs="1.1" Color="d0d0d0"/>
```

In the Palace config file, this results in an Impedance boundary with Rs value mapped to the surface.

```
"Impedance": [
    {
        "Attributes": [
            37,
            38
        ],
        "Rs": 0.35
    },
    {
        "Attributes": [
            39
        ],
        "Rs": 1.1
    }
],
```

15.3 Extended XML stackup format

The XML stackup format has grown beyond the basic dielectric stack and conductor/via layer mapping described above. The current major additions, all optional and backward-compatible with older stackup files, are:

- **Derived layers** — a layer's geometry can be computed from other layers via boolean operations (`AND`/`OR`/`XOR`/`NOT`) and resizing, instead of only being read directly from GDSII. Used, for example, to derive on-chip resistor geometry from existing process layers.
- **Reference-relative positioning** — a `<Layer>` or `<Dielectric>` can be positioned as an offset from the top or bottom edge of another named layer/dielectric, instead of requiring an absolute Zmin/Zmax.
- **Thermal material properties** — `Density`, `ThermalConductivity` and `ThermalConductivityTable` on `<Material>`, plus a `<Tables>` block for temperature-dependent conductivity, used by the Elmer thermal flow described in chapter "Using gds2palace with Elmer FEM for thermal simulation".

Using derived layers, reference-relative positioning, or thermal conductivity tables requires `schemaVersion="3.0"` in the `<Stackup>` root element. The stackup reader prints a warning if a file declares a newer `schemaVersion` than the installed gds2palace version supports, so an outdated installation is easier to notice.

The full attribute reference, with examples, is in `XML_stackup_format.md`, and derived-layer details specifically are in `derived_layers.md`.

15.4 Software versions used in this document

Palace version used for this document: v0.14
Palace installation method used: container for Apptainer environment, running on Ubuntu 24.04.02 LTS
Simulation host: AMD Ryzen 9-7950X (16 cores) with 128 GB RAM
Code base used for this test: gds2palace Python scripts as of October 13, 2025 (unless noted otherwise)

16 List of examples

Here is an overview of model examples.

16.1 palace_line_viaport.py

Microstrip line on TopMetal2 over Metal1 ground plane, with via ports on both ends. Geometry is read from GDSII. The stackup does not include bulk silicon because that is shield by the ground plane anyway.

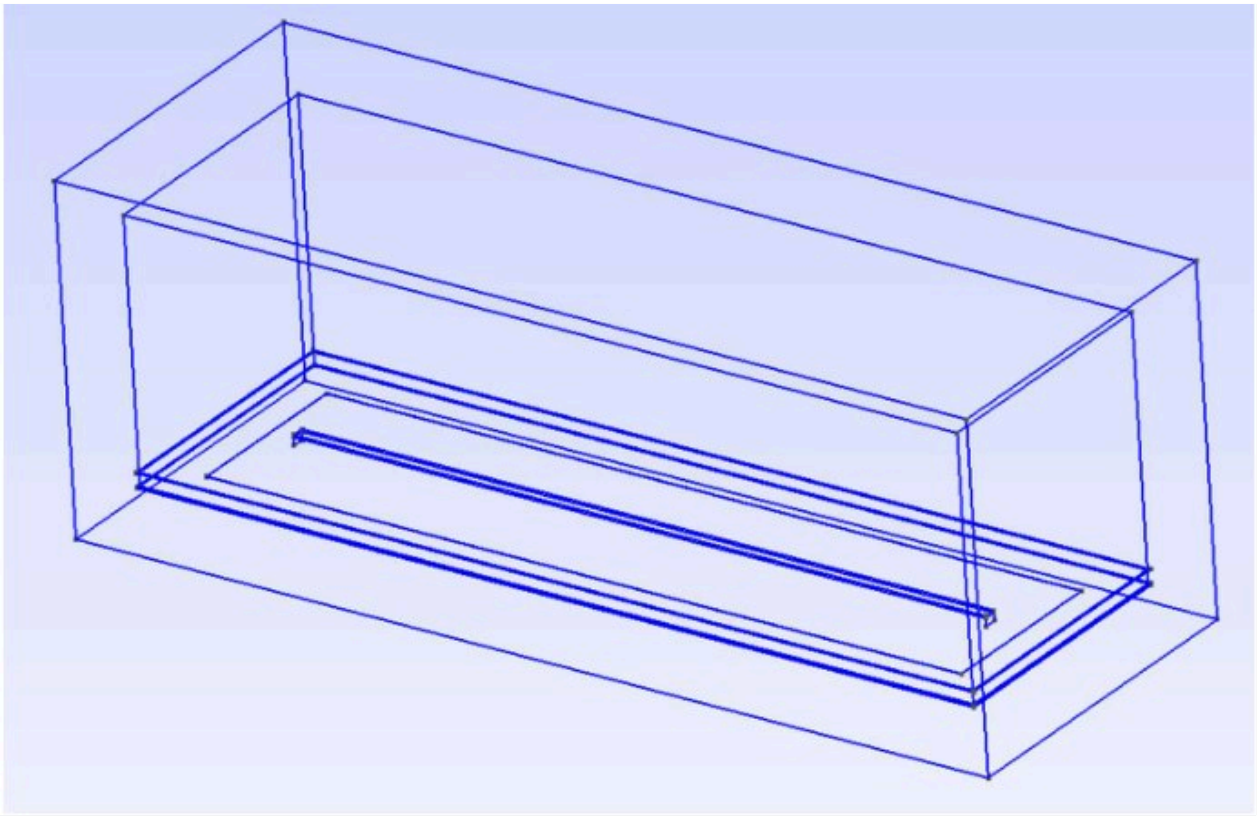


Figure 38: Example palace_line_viaport.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_line_viaport.py

16.2 palace_line_noGDS.py

Similar to palace_line_viaport.py but geometry is created by code instead of reading it from GDSII. https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_line_noGDS.py

16.3 palace_ind_frame.py

This is a 2-port inductor embedded into a metal frame. Geometry from GDSII.

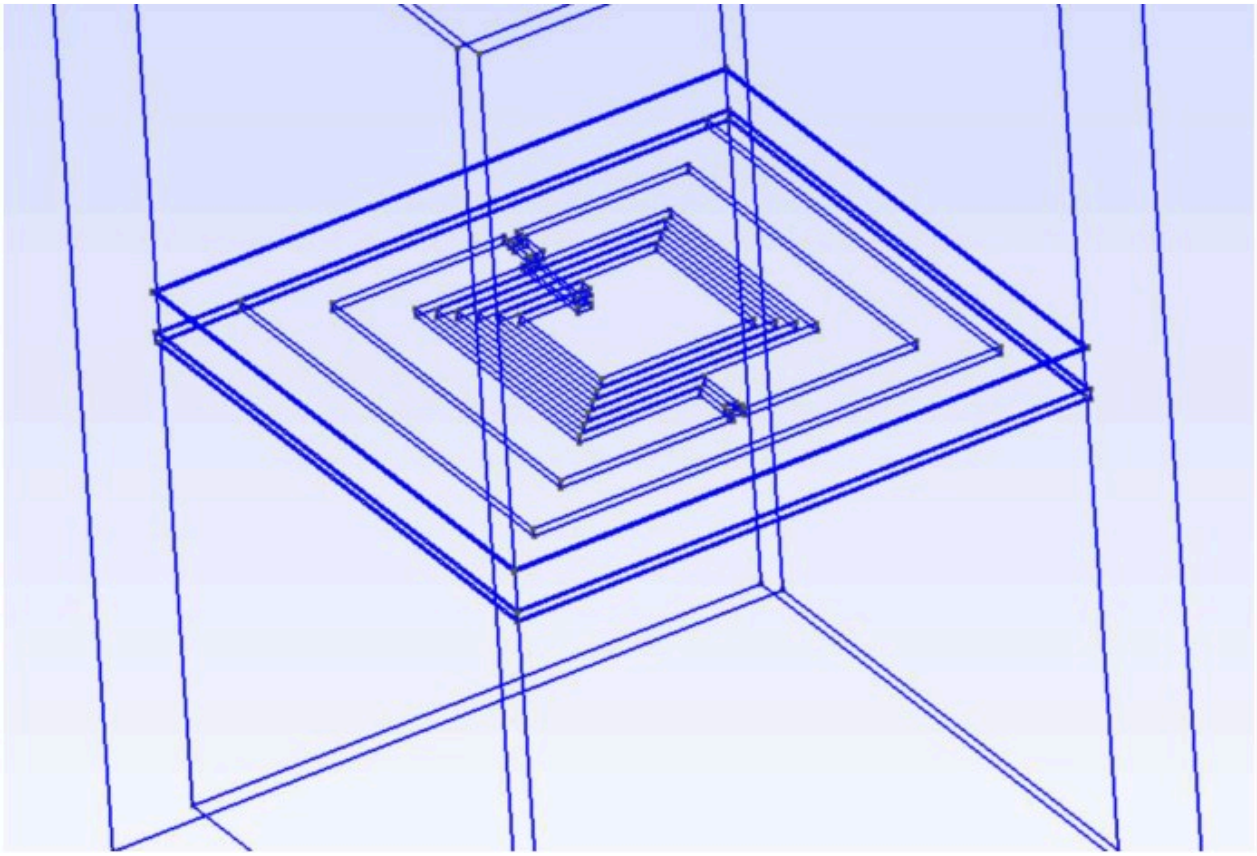


Figure 39: Example palace_ind_frame.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_ind_frame.py

16.4 palace_L2n0.py

2-port octagon inductor with via ports down to an artificial metal plane that sits on the surface of bulk silicon.

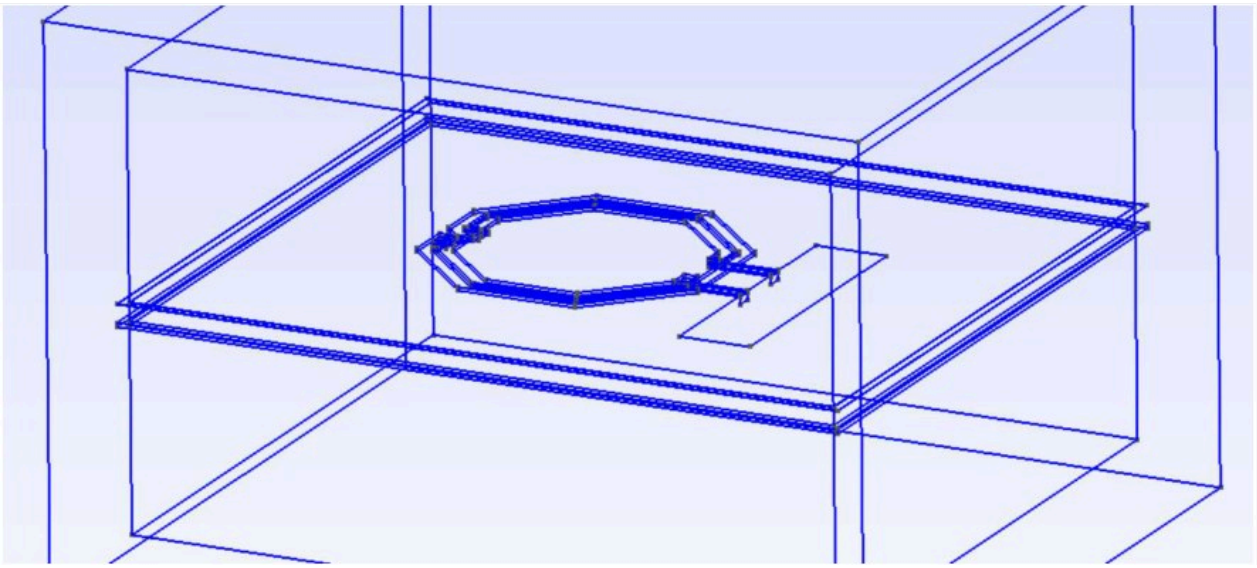


Figure 40: Example palace_L2n0.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_L2n0.py

Field dump at a single frequency is enabled in this model. The Paraview screenshot below shows current density using this file:

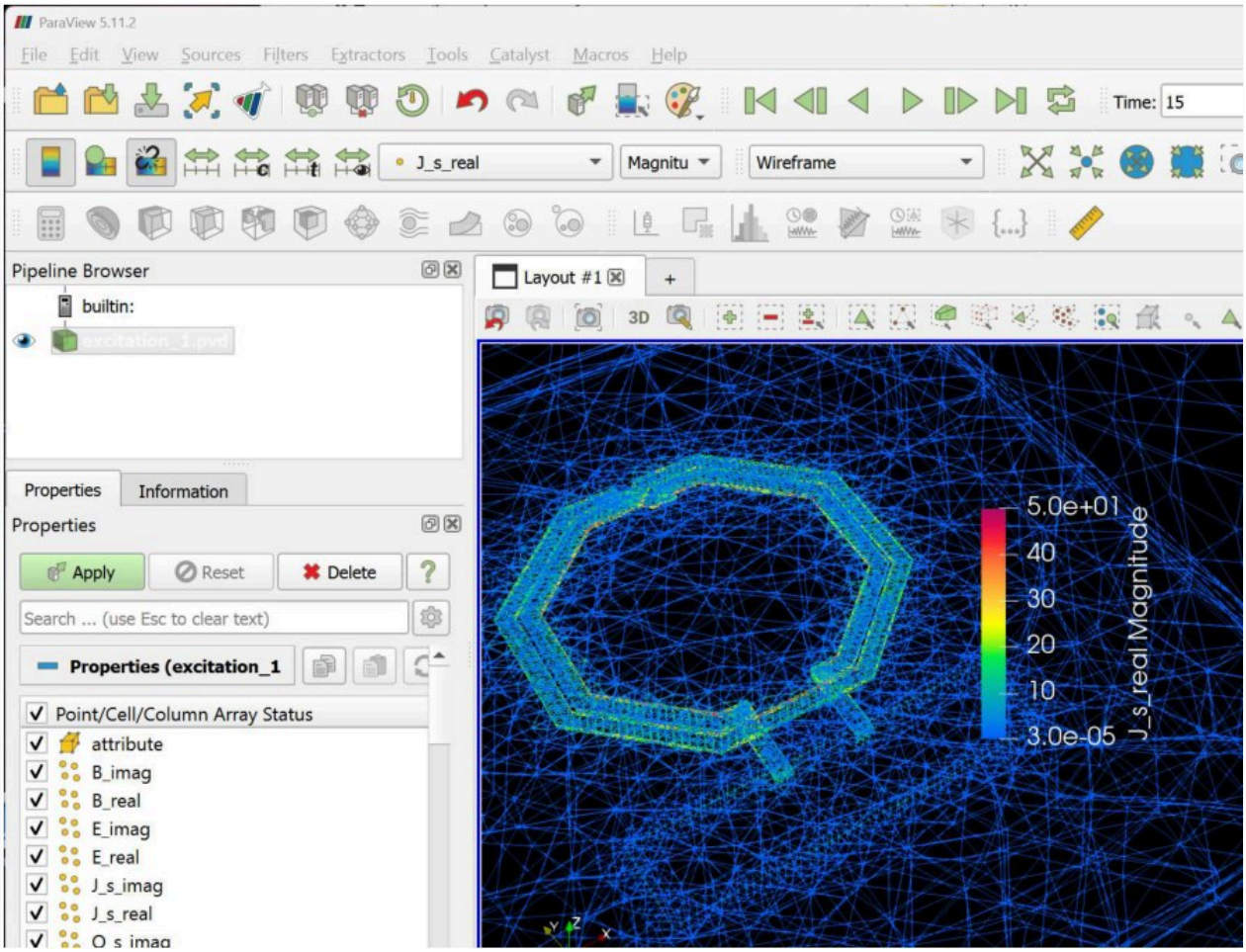


Figure 41: Example palace_L2n0.py: current density in ParaView

16.5 palace_butlermatrix.py

This is quite a big model, the Butler Matrix by Ardavan Rahimian in IHP Open PDK Tapeout - July 2025, available at https://github.com/IHP-GmbH/TO_July2025/tree/main/W_Band_Butler_Matrix_IC

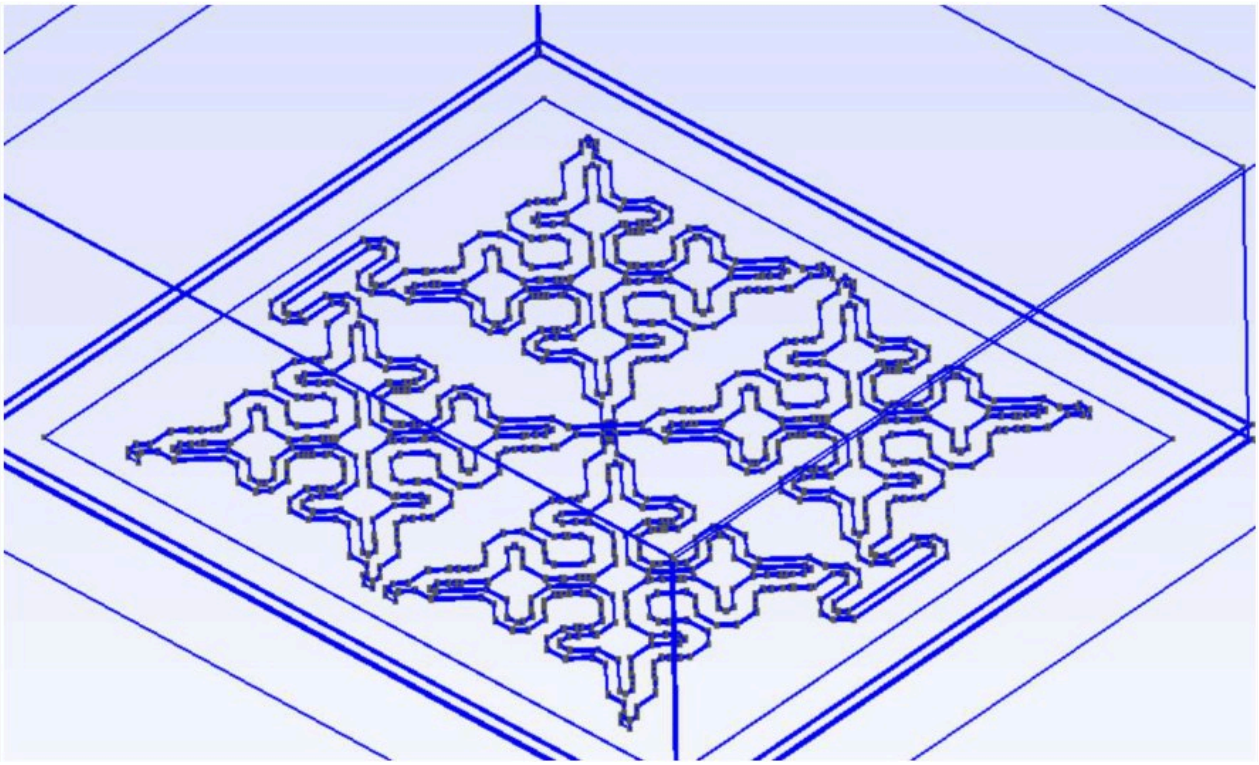


Figure 42: Example palace_butlermatrix.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_butlermatrix.py

Geometry is from GDSII, the model uses a total of 8 via ports between Metal3 and TopMetal2. However, to speed up simulation, only one port excitation is active in this model, and we only get S11,S21,S31,S41,S51,S61,S71 and S81 from this simulation run.

```
simulation_ports = simulation_setup.all_simulation_ports()
# instead of in-plane port specified with target_layername,
# we here use via port specified with from_layername and to_layername
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=1, voltage=1, port_Z0=50, source_layernum=201,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=2, voltage=0, port_Z0=50, source_layernum=202,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=3, voltage=0, port_Z0=50, source_layernum=203,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=4, voltage=0, port_Z0=50, source_layernum=204,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=5, voltage=0, port_Z0=50, source_layernum=205,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=6, voltage=0, port_Z0=50, source_layernum=206,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=7, voltage=0, port_Z0=50, source_layernum=207,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=8, voltage=0, port_Z0=50, source_layernum=208,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
```

All other port excitations are set to voltage=0 in the model. To get full 8-port S-parameters, we would need to change all port voltages to 1 and re-run the model, so that all excitations are simulated, one after another.

16.6 palace_butlermatrix_dump93.py

Same as before, but a field dump setting is added in this model at 93 GHz.

```
settings['fstart']  = 90e9
settings['fstop']   = 95e9
settings['fstep']   = 0.1e9

# settings["fpoint"] = [93e9]  # list of discrete frequencies to be simulated

settings["fdump"] = [93e9]  # save field dump at these frequency points
```

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_butlermatrix_dump93.py

16.7 palace_core.py

This is the 4-port “core” model of the 60 GHz medium power amplifier from IHP Analog Academy online course. There are two via ports at the input and output, and two in-plane ports on Metal2 between the common emitter polygon and base/collector coming down from the via stacks.

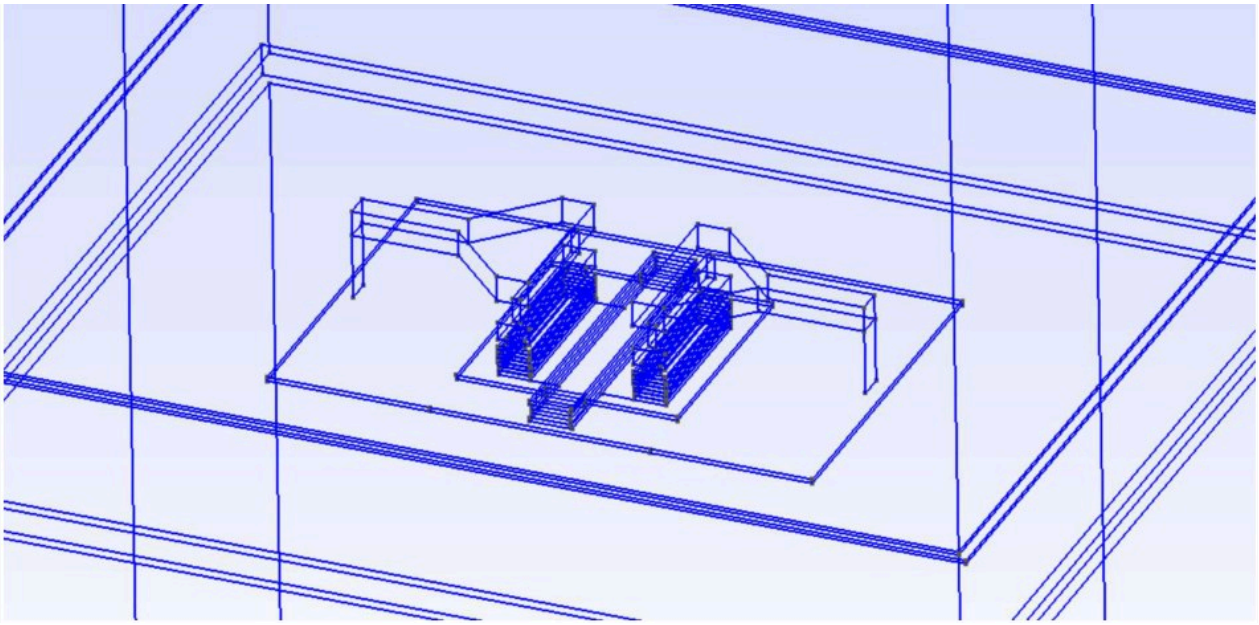


Figure 43: Example palace_core.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_core.py

16.8 palace_rfcmim.py

This is 2-port model of an rf_cmim component with some feedline length. Via ports are used at the input and output down to the Metal1 ground plane. Via arrays are merged by this setting: merge_polygon_size = 2

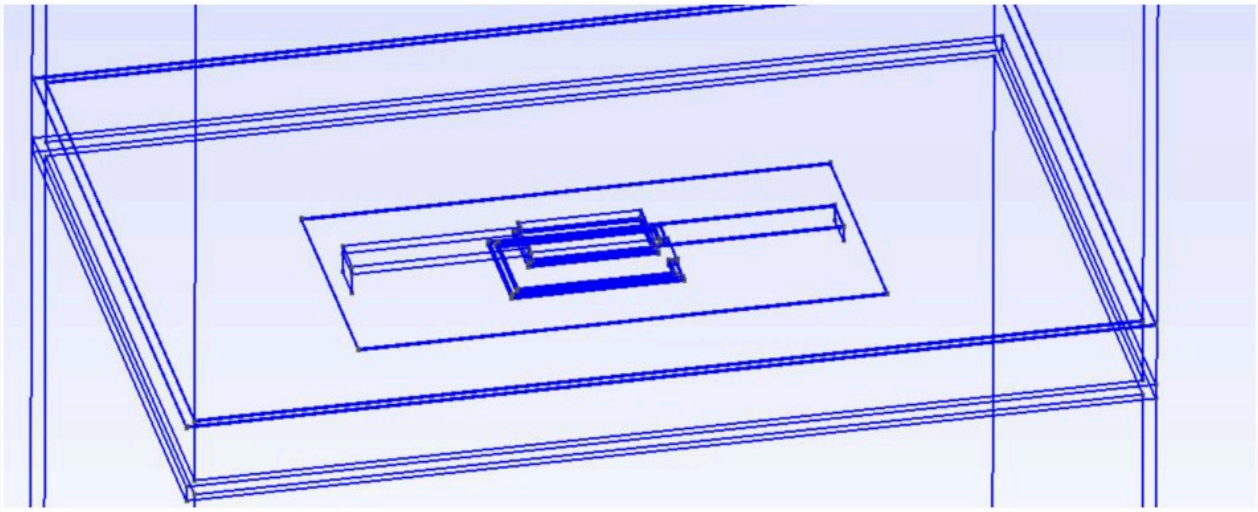


Figure 44: Example palace_rfcmim.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_rfcmim.py

16.9 palace_pcb_lowpass.py

This shows a use case of gds2palace outside the RFIC domain: the layout of a PCB lowpass on RO4003 substrate was imported to klayout and saved in GDSII format.

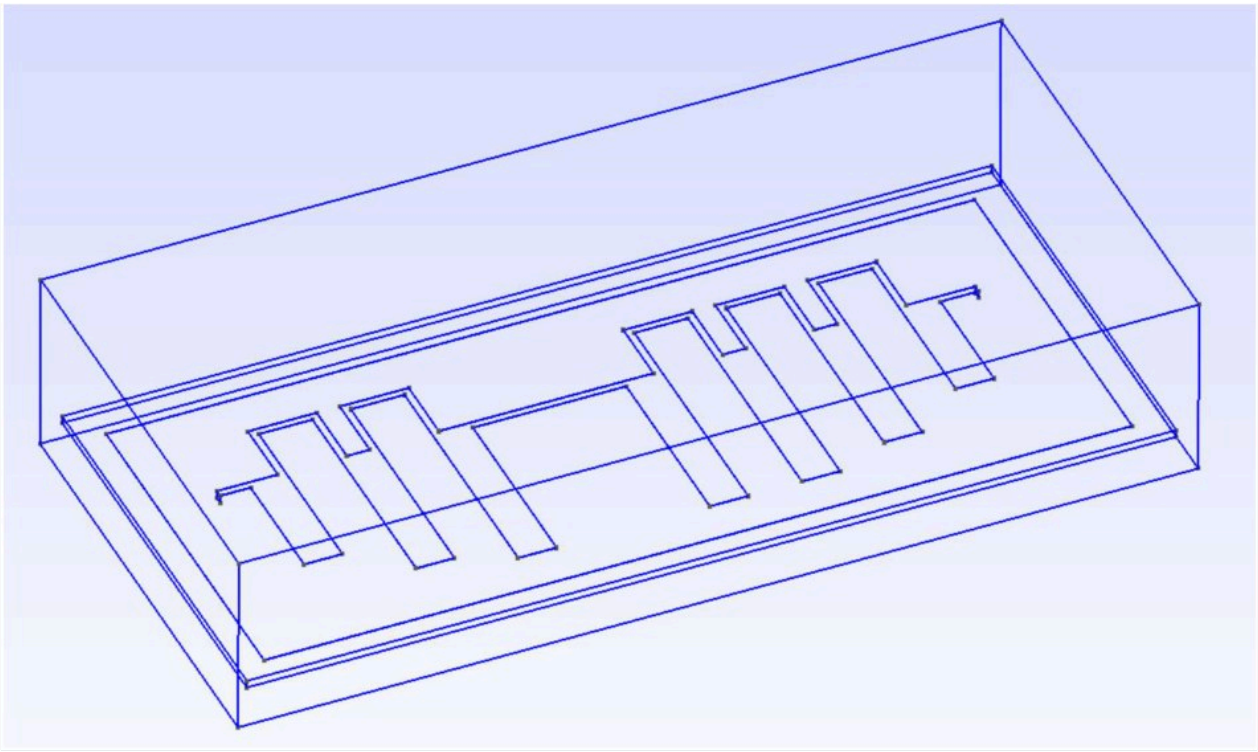


Figure 45: Example palace_pcb_lowpass.py

An XML stackup file was created for this substrate, with layout layers matching the layer numbers used in klayout.

```
<Stackup schemaVersion="2.0">
  <Materials>
    <Material Name="Copper" Type="Conductor" Permittivity="1" DielectricLossTangent="0" Conductivity="3.3e7"
Color="00ff00"/>
    <Material Name="RO4003" Type="Dielectric" Permittivity="3.38" DielectricLossTangent="0.0022"
Conductivity="0" Color="01e0ff"/>
    <Material Name="AIR" Type="Dielectric" Permittivity="1.0" DielectricLossTangent="0.0" Conductivity="0"
```

```
Color="d0d0d0"/>
  </Materials>
  <ELayers LengthUnit="um">
    <Dielectrics>
      <Dielectric Name="RO4003" Material="RO4003" Thickness="510"/>
    </Dielectrics>
    <Layers>
      <Substrate Offset="0"/>
      <Layer Name="Bottom" Type="Conductor" Zmin="-17" Zmax="0" Material="Copper" Layer="1"/>
      <Layer Name="Top" Type="Conductor" Zmin="510" Zmax="527" Material="Copper" Layer="10"/>
    </Layers>
  </ELayers>
</Stackup>
```

Here, setting “margins” only controls the (small) oversize of the dielectrics from the drawing layers. The optional “air_around” setting was used in the simulation model to define the air margins independently, without adding these air layers in the XML stackup.

```
settings['unit']   = 1e-6  # geometry is in MILLIMETER for PCB example
settings['margin'] = 2000    # distance in microns from GDSII geometry boundary to simulation boundary
settings['air_around'] = [1000,1000,1000,1000,2000, 10000] # airbox size to simulation boundary
```

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_pcb_lowpass.py